AMERICAN UNIVERSITY OF BEIRUT

Final Year Project – Spring Semester: AI Detection for Cognitive Impairment – Alzheimer's Disease

by

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Executive Summary

Motivation for the Project: Globally, the frequency of cognitive impairment disorders such as Alzheimer's disease and related dementias continues to rise, placing an increasing emotional and financial burden on patients, families, and healthcare systems. Current diagnostic practices rely heavily on clinical tests that are invasive, time-consuming, and expensive, often delaying early intervention. This motivates the development of automated, accessible, and non-invasive tools capable of detecting subtle linguistic and acoustic biomarkers of cognitive decline using simple speech samples. This project addresses this need by developing an interpretable machine learning–based speech analysis system that assists clinicians and researchers in preliminary cognitive screening.

Problem Definition: The aim of this project is to develop a data-driven, machine learning–powered framework that analyzes linguistic, acoustic, semantic, and biomarker-based features to differentiate between cognitively healthy control participants and participants diagnosed with dementia. The system processes natural speech—such as picture description, story recall, verbal fluency, and sentence construction—to identify patterns in word choice, fluency, timing, prosody, and semantic content that correlate with cognitive status. A major challenge lies in obtaining high-quality annotated speech data and designing preprocessing pipelines that can reliably handle noise, variable recording quality, corrupted audio files, class imbalance, and transcript inconsistencies.

Solution Approach: Our solution integrates natural language processing (NLP), speech signal processing, semantic feature extraction, and machine learning classification techniques. We extracted a range of linguistic, acoustic, semantic, and biomarker-based features from speech recordings and CHAT transcripts in the DementiaBank Pitt Corpus. The final enhanced feature set contains 32 variables, including syntactic complexity, pronoun-noun ratio, repetition rate, pause duration, pitch variability, jitter, shimmer, information-unit coverage, story-recall similarity, disfluency rates, and task-normalized z-scores. These features were then normalized and used to train and compare several models, including Support Vector Machines, Logistic Regression, Random Forests, XGBoost, BERT-based models, and ensemble approaches. The system was developed with reproducibility in mind, using structured preprocessing pipelines and modular training scripts.

Constraints and Challenges: The project operates under several key constraints. First, the dataset size is inherently limited due to privacy restrictions and the sensitive nature of clinical speech recordings. This affects model generalization and requires careful feature selection, participant-level splitting, and class-imbalance handling. Second, ethical and procedural constraints apply, as working with medical data requires Institutional Review Board (IRB) compliance. Our work with the DementiaBank Pitt Corpus adheres to confidentiality and usage restrictions, with clinical guidance from collaborators. Third, variability in audio quality, inconsistent transcript formatting, corrupted WAV headers, and missing modalities introduce preprocessing complexity. These technical constraints necessitate robust cleaning procedures, error-tolerant feature extraction methods, and the exclusion or repair of problematic files. Finally, computational constraints arise when training and deploying heavy transformer-based models, so the final design prioritized interpretable handcrafted features and efficient classical machine learning models that can run on commodity hardware.

Main Findings: The team completed the literature review, finalized system requirements, and implemented the complete data acquisition, preprocessing, feature extraction, model training, evaluation, interpretability, and prototype development pipeline. Collaboration was established with medical advisors to ensure clinical relevance. The final cleaned dataset contained 1,107 valid samples after audio quality control and feature cleaning. Several modeling strategies were evaluated under a participant-level train, validation, and test split to prevent patient leakage. The best-performing system was a task-routed ensemble that uses separate specialists for Cookie Theft and non-Cookie tasks. This final model achieved 92.2% test accuracy and a macro-F1 score of 0.855 on the held-out test set, outperforming the baseline classical models, BERT-only models, and the simple averaging ensemble. SHAP analysis showed that information-unit coverage was the most influential feature, followed by word count, task-normalized syntactic and semantic features, disfluency biomarkers, and acoustic features such as pitch variability and intensity range.

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Chapter 1

Introduction

One of the most important public health challenges of the twenty-first century is cognitive impairment, especially Alzheimer's disease (AD). The World Health Organisation (WHO) currently reports that an estimate of 10 million new cases of dementia are diagnosed annually, with over 55 million people worldwide suffering from the disease [1]. Furthermore, about 60–70% of dementia cases are caused by Alzheimer's disease alone, which puts a huge emotional, social, and financial strain on families and healthcare systems. Hence, early detection methods are imperative as AD is highly progressive and irreversible. As a result, early diagnosis and treatment can improve patients' overall quality of life, alleviate carer stress, and delay functional decline [2].

Current diagnostic methods rely primarily on laboratory biomarkers, clinical interviews, and neuroimaging, all of which are not only time-consuming but can be costly, and invasive [3]. These techniques are unavailable in many low- and middle-income areas due to their reliance on advanced medical infrastructure and specialized care. Furthermore, in the early stages of Alzheimer's disease, subtle cognitive changes often go undetected, increasing the rate of false negatives and decreasing the effectiveness of therapeutic outcomes [4]. Research into low-cost, non-invasive and automated tools that can help clinicians detect cognitive decline earlier and more objectively has been instigated by this widening diagnostic gap.

Recent developments in Natural Language Processing (NLP) and Artificial Intelligence (AI) have shown that a person's speech patterns can reveal neurological abnormalities connected to AD [5]. This includes variations in lexical diversity, syntactic complexity, speech rate, pause frequency, and semantic coherence which in specific has been found to be among the most accurate linguistic biomarkers of cognitive impairment [3–5]. Patients with AD tend to speak more slowly and fragmentedly, have a smaller range of vocabulary, and have trouble staying on topic as their memory and executive functions decline with the progression of the disease [6]. With spontaneous speech involving quantifiable linguistic cues, using speech patterns to provide a powerful and non-invasive insight into cognitive health can be incredibly valuable.

Using this inspiration, this project developed an AI-based speech analysis system for early Alzheimer's detection. The system extracts 32 enhanced acoustic, linguistic, semantic, biomarker-based, and task-normalized features that reflect cognitive performance from the DementiaBank Pitt Corpus [7], which includes audio recordings and transcripts from clinically diagnosed patients and healthy controls. To categorise samples as Control or Dementia, machine-learning models process these variables, which include pause duration, speech rate, syntactic complexity, disfluency rates, information-unit coverage, story-recall similarity, and acoustic voice-quality measures. The developed system seeks to provide a scalable, non-invasive, language-based screening tool for early cognitive assessment by fusing automated feature extraction with clinically validated tasks, including picture description, story recall, verbal fluency, and sentence construction.

The final implementation includes transcript preprocessing, audio feature extraction, semantic feature extraction, participant-level data splitting, model comparison, task-routed ensemble classification, SHAP-based interpretability, and a prototype web interface. The long-term objective is to make digital screening easily accessible in clinical and residential settings, assisting neurologists in monitoring the course of the disease and enhancing the results of early intervention.

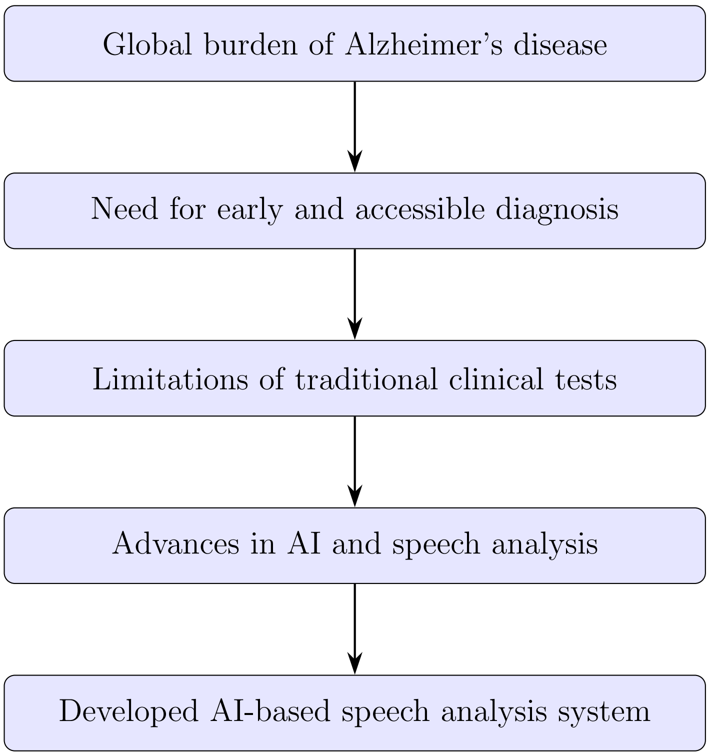


Figure 1.1: Motivation for developing a non-invasive AI-based system for early Alzheimer's detection.

1.1 Motivation and Desired Needs

The urgent need for easily accessible, early, and non-invasive Alzheimer's disease diagnostic tools is what inspired this project. The prevalence of dementia is rising at a startling rate as the world's population grows. Nearly 70% of dementia cases worldwide are caused by Alzheimer's disease alone, and millions of new cases are diagnosed with it annually [1, 2]. Due to the irreversible nature of the disease, early detection is essential to enable timely therapeutic interventions and lower the socioeconomic burden on families and healthcare systems.

Even though accurate diagnostic techniques exist, such as MRIs, PET scans, and cerebrospinal fluid (CSF) analyses, these are usually costly, invasive, and unsuitable for high-throughput screening. Speech-based analysis, on the other hand, provides a more comfortable, affordable, and approachable substitute. Studies have repeatedly demonstrated that subtle but measurable linguistic changes are indicative of cognitive impairment. This includes decreased semantic coherence, syntactic simplification, increased pauses, and decreased vocabulary richness [3–5]. Hence, the use of AI-based linguistic analysis systems as effective instruments for cognitive screening through natural speech is supported by these findings.

By developing an AI-based system that automatically extracts and analyses linguistic, acoustic, semantic, and disfluency patterns from spontaneous speech, this project aims to address these issues. The final model computes 32 enhanced quantitative features across four cognitive tasks—**picture description**, **story recall**, **verbal fluency**, and **sentence construction**—by processing recordings from standardised clinical tasks in the DementiaBank dataset. Language fluency, memory recall, and semantic organization—three of the first areas impacted by Alzheimer's disease—are assessed by these tasks taken together.

Project Objectives

The primary objectives of this project are:

- To develop an automated speech-based AI system capable of distinguishing between Control and Dementia speech samples.
- To design and implement a robust feature extraction pipeline integrating acoustic, linguistic, semantic, biomarker-based, and task-normalized markers.
- To evaluate the system's predictive performance using classical machine-learning algorithms, XGBoost, BERT-based models, and ensemble approaches.
- To develop a prototype web interface that demonstrates how the system can support remote, speech-based cognitive screening using only audio and transcript input.

Desired Needs

The proposed system addresses several essential needs across medical, technical, ethical, and societal dimensions:

- **Medical Need:** Provide researchers and clinicians with a non-invasive, affordable, and reliable screening tool that uses speech analysis to identify cognitive impairment. The system supports evidence-based decision-making and improves conventional assessments by providing unbiased and data-driven insights.

- **Technical Need:** Establish a strong AI pipeline that can extract, normalise, and analyse linguistic, acoustic, semantic, and biomarker-based features from unplanned speech, including semantic coherence, syntactic complexity, pause duration, pitch variability, and disfluency rates. To find early signs of Alzheimer's, the pipeline combines machine learning classification, feature computation, and NLP preprocessing.

- **Ethical and Privacy Need:** Ensure complete adherence to international data-protection regulations (HIPAA, GDPR) and the American University of Beirut Institutional Review Board (IRB). To ensure participant confidentiality, the project only uses unidentified and ethically sourced data from the DementiaBank Pitt Corpus [7].

- **Societal Need:** Replace expensive neuroimaging procedures with automated speech analysis to increase access to early cognitive screening in remote and low-resource areas. This strategy reduces the stigma related to cognitive testing, aids public health initiatives focused on early intervention, and gives clinicians readily available, technologically advanced diagnostic tools.

In conclusion, the goal of this project is to develop a scalable, interpretable, and ethical system for early Alzheimer's detection in an effort to close the gap between clinical neuroscience and artificial intelligence. The system is a useful step towards a fair, data-driven evaluation of cognitive health by coordinating clinical goals with technological advancement.

Chapter 2

Design Process

This chapter describes the design process followed throughout the development of the final system. It begins by identifying the technical and non-technical constraints that shaped the project, then discusses the design alternatives considered, the final design decisions, the main design iterations, and the standards that guided the implementation.

2.1 Engineering Constraints

The design of the proposed system was directly affected by dataset limitations, clinical-data restrictions, computational resources, audio quality, model interpretability, and deployment requirements. Since the project deals with speech-based cognitive screening, the design process had to balance model accuracy with ethical safety, reliability, reproducibility, and practical implementation within the two-semester FYP timeline.

2.1.1 Technical Constraints

1. **Data Limitations:** The main dataset used in this study is the DementiaBank Pitt Corpus [7]. This dataset contains English speech recordings and CHAT transcripts, which limits the system's ability to generalize to speakers of other dialects or languages such as Arabic, the primary language for many potential users in Lebanon and the Middle East. The dataset is also relatively small for deep learning applications and is strongly imbalanced, with substantially more Dementia samples than Control samples. This required careful feature selection, participant-level splitting, and class-imbalance handling during model training.

2. **Speech and Audio Processing Complexity:** The system depends heavily on accurate extraction of linguistic, acoustic, semantic, and biomarker-based features. Even small differences in audio quality, microphone condi-

tions, silence regions, or file formatting can affect features such as pause duration, pitch variability, jitter, shimmer, and speech-time ratio. During implementation, some WAV files also had invalid or corrupted headers, which required repairing, re-encoding, or excluding problematic files before feature extraction.These acoustic and linguistic markers have been widely used in previous speech-based Alzheimer's detection studies [3, 4, 8].

3. **Transcript Formatting and CHAT Annotation Complexity:** The transcripts in the DementiaBank Pitt Corpus follow the CHAT format, which contains participant lines, investigator lines, annotations, retracing markers, repetition symbols, pause symbols, and filled-pause tokens. These elements had to be processed carefully because some markers are noise, while others are clinically useful. For example, filled pauses such as "uh" and "um" were preserved because they provide useful disfluency information for cognitive-impairment analysis.

4. **Server and Computing Requirements:** Training large transformer-based or end-to-end deep learning models requires significant computational resources. Due to student-level hardware and limited access to persistent GPU resources, the final design had to remain computationally feasible. This constraint influenced the decision to prioritize handcrafted speech features, efficient classical machine learning models, and ensemble methods that can run on commodity hardware.

5. **Integration and Latency:** The system includes several interconnected modules, including transcript preprocessing, audio preprocessing, feature extraction, model inference, prediction explanation, and the web interface. Heavy models can increase inference time and make deployment less practical. Therefore, the final system was designed to remain lightweight enough for fast prediction while still providing interpretable results.

6. **Model Generalization and Evaluation:** Since multiple recordings can belong to the same participant, a normal random split could cause data leakage if samples from the same speaker appear in both training and testing. To avoid this, the final evaluation used participant-level train, validation, and test splits. This ensured that the model was evaluated on unseen participants rather than simply recognizing patterns from a known speaker.

7. **Project Timeline and Resource Limitations:** The project was completed by three undergraduate students over the FYP period while managing other academic responsibilities. The scope therefore had to focus on the most important deliverables: data preprocessing, feature extraction, model training, evaluation, interpretability, and a working prototype interface.

2.1.2 Non-Technical Constraints

1. **Ethical and Privacy Considerations:** Since the project involves clinical speech data, ethical and privacy considerations were central to the design process. The system uses anonymized data from the DementiaBank Pitt Corpus and follows the dataset's usage restrictions. The model is intended strictly as a research and screening tool and must not be interpreted as a replacement for clinical diagnosis.

2. **Health and Safety:** Because the system deals with cognitive health, the output must be communicated carefully. A false prediction could cause unnecessary anxiety or false reassurance. Therefore, the final system is designed to provide a screening probability and explanation, not a medical diagnosis. Clear disclaimers are required to prevent misuse.

3. **Economic Constraint:** Commercial cloud computing, large-scale storage, and GPU-based deployment may exceed a student project budget. This constraint encouraged the use of open-source tools, local processing, and efficient machine learning models that do not require expensive infrastructure.

4. **Social and Cultural Constraints:** The current dataset is English-based and mainly reflects North American speech patterns. This limits the system's inclusiveness for Arabic-speaking or bilingual users in Lebanon and the wider region. Future versions would need Arabic speech data, local validation, and multilingual language-processing support.

5. **Regulatory Constraint:** Any future clinical use would require institutional approval, medical validation, and compliance with health-data regulations. Therefore, the current prototype is positioned as an academic research system and not as an approved medical device.

6. **Sustainability and Maintenance:** The project depends on several software libraries for audio processing, natural language processing, machine learning, and web deployment. Long-term use would require maintenance, library updates, and model retraining if new data become available.

2.2 Design Alternatives

Several design alternatives were considered before selecting the final architecture. Each alternative was evaluated based on accuracy, interpretability, computational feasibility, robustness, and suitability for the available dataset.

Design Alternative 1: Classical Machine Learning on Handcrafted Features

The first design alternative relied on handcrafted acoustic, linguistic, and semantic features extracted from audio recordings and transcripts. CClassical models such as Logistic Regression, Support Vector Machines, Random Forests, and XGBoost were considered because they are suitable for small-to-medium clinical datasets and provide a higher level of interpretability than end-to-end neural networks [3, 4, 9].

This approach was attractive because it allowed the system to connect model decisions to clinically meaningful speech markers such as pauses, repetitions, syntactic complexity, semantic coherence, information-unit coverage, jitter, and shimmer. However, using one classical model alone was not sufficient to fully capture task-specific differences across Cookie Theft, recall, fluency, and sentence-construction samples.

Design Alternative 2: Deep Learning and Transformer-Based Models

The second design alternative considered deep learning and transformer-based architectures, including BERT-based text models and multimodal fusion approaches [6, 10, 11]. These models can learn complex representations from transcripts and audio and have shown strong performance in recent literature.

However, the dataset size and compute limitations made heavy transformer-based training less practical. The available number of training samples was limited, and the risk of overfitting was high. In addition, transformer-based models are less interpretable and more difficult to deploy efficiently on commodity hardware. Although BERT-based experiments were useful for comparison, they did not become the final selected design.

Design Alternative 3: Simple Averaging Ensemble

The third alternative combined the predictions of multiple classical models using simple probability averaging. This design improved robustness compared to using a single classifier because each model captured different decision patterns. For example, XGBoost handled nonlinear feature interactions, Random Forest provided tree-based robustness, SVM captured nonlinear separation, and Logistic Regression provided a simple linear baseline.

This approach improved the results, but it still treated all cognitive tasks in the same way. This became a limitation because some features are meaningful only for specific tasks. For example, information-unit coverage is highly useful for the Cookie Theft picture-description task, but it is not equally meaningful for fluency, recall, and sentence-construction tasks.

Final Selected Alternative: Task-Routed Ensemble

The final selected design was a task-routed ensemble. In this architecture, samples are first routed based on their task type. Cookie Theft samples are sent to a Cookie specialist ensemble, while the remaining tasks are sent to a non-Cookie specialist ensemble. This allowed the model to use task-specific information more effectively, especially because the strongest semantic features are not equally relevant across all task types.

The task-routed ensemble was selected because it achieved the best final performance while remaining interpretable, efficient, and compatible with the available dataset and deployment constraints.

2.3 Design Decisions

Based on the constraints and alternatives above, several key design decisions were made.

1. **Using the DementiaBank Pitt Corpus as the primary dataset:** The Pitt Corpus was selected because it provides both speech recordings and CHAT transcripts from standardized cognitive tasks. This made it suitable for a multimodal speech-analysis pipeline.

2. **Using participant-level data splitting:** Since participants can have multiple task recordings, the data were split by participant rather than by individual sample. This decision prevented data leakage and ensured that the test set contained unseen participants.

3. **Using enhanced handcrafted features instead of relying only on deep embeddings:** The final system uses 32 enhanced features across linguistic, acoustic, semantic, biomarker, and task-normalized categories. This decision improved interpretability and worked better with the available clinical dataset size.

4. **Using Parselmouth/Praat for acoustic feature extraction:** Acoustic extraction was moved toward Parselmouth/Praat because it was much faster and more practical than the earlier Librosa-based approach. This reduced extraction time significantly and made repeated experimentation possible.

5. **Preserving clinically useful disfluency markers:** Filled pauses, repetitions, revisions, and fragments were treated as useful biomarkers rather than simply removed as noise. This allowed the model to capture speech behaviors associated with cognitive decline.

6. **Using class-imbalance handling:** Since the dataset contained many more Dementia samples than Control samples, imbalance handling was necessary. The final design used class-weighting strategies because they preserved the real data distribution better than synthetic oversampling approaches such as SMOTE.

7. **Using task routing:** Task routing was selected because the four cognitive tasks do not produce the same type of speech information. Separating Cookie Theft from non-Cookie tasks allowed the model to make better use of task-specific features.

8. **Using SHAP for interpretability:** SHAP analysis was selected to explain global and per-sample model behavior, following the need for interpretability in medical AI screening systems [6, 12]. This was important because medical AI systems must be interpretable and defensible, especially when used for screening support.

2.4 Design Iterations

The final system was reached through several design iterations. These iterations were essential because early versions of the pipeline revealed practical issues in preprocessing, feature extraction, model performance, and deployment feasibility.

1. **Transcript-cleaning iteration:** In an early version of the transcript-cleaning pipeline, filled pauses such as “uh” and “um” were removed during cleaning. This was later corrected because filled pauses are diagnostic speech markers and should be preserved as disfluency biomarkers.

2. **Audio-quality iteration:** During acoustic feature extraction, several WAV files were found to have invalid or corrupted headers. A repair step was introduced using FFmpeg-based conversion where possible, and unrecoverable files were excluded from the final clean dataset.

3. **Acoustic-extraction optimization:** The first acoustic extraction approach relied heavily on slower audio-processing methods. The pipeline was later optimized using Parselmouth/Praat, which made feature extraction much faster and allowed more efficient experimentation.

4. **Feature-cleaning iteration:** Initial features included redundant or weak variables. After exploratory analysis, some features were removed due to high correlation or low statistical signal, while stronger biomarker rates and task-normalized features were added. This led to the final enhanced feature set.

5. **Modeling iteration:** The team first tested individual baseline models, then XGBoost, then BERT-based models and fusion experiments, and finally ensemble models. The results showed that handcrafted domain-specific features were more effective than heavier transformer-based models for this dataset size.

6. **Task-routing iteration:** A single global model was not ideal because different tasks produce different feature meanings. The final iteration introduced task routing, where Cookie Theft samples and non-Cookie samples were handled by separate specialist ensembles. This became the final selected architecture.

7. **Interpretability iteration:** After selecting the final model, SHAP analysis was added to identify the most influential features and support explainable prediction. This helped validate the clinical relevance of the model's decisions.

2.5 Standards

The project design and development followed recognized engineering, software, privacy, and speech-processing standards to ensure technical reliability, safety, and responsible use.

- **IEEE 830-1998 – Software Requirements Specification:** This standard guided the documentation of system requirements, including functional and non-functional requirements such as classification output, reproducibility, latency, and interpretability [13].

- **IEEE 12207 – Software Life Cycle Processes:** This standard relates to the planning, design, implementation, testing, and maintenance of the software pipeline. The project followed a staged development process that included preprocessing, feature extraction, model training, testing, and deployment [14].

- **ISO/IEC 27001 – Information Security Management:** This standard is relevant because the project deals with clinical speech data. It supports the need for secure data handling, anonymization, restricted access, and responsible storage of sensitive information [15].

- **HIPAA – Health Insurance Portability and Accountability Act:** HIPAA principles guided the ethical handling of clinical information. Although the project uses anonymized research data, the system is designed with privacy and confidentiality in mind [16].

- **GDPR – General Data Protection Regulation:** GDPR principles are relevant for consent, data minimization, privacy rights, and the responsible handling of personal data. These principles are important for any future version that collects speech from real users [17].
- **ITU-T P.800 – Speech Quality Assessment:** This standard is relevant to the speech-processing aspect of the project because acoustic quality, speech clarity, and voice-related measurements affect the reliability of extracted features [18].

Overall, these standards supported the design of a system that is not only technically functional, but also reproducible, interpretable, privacy-aware, and suitable for responsible research use.

Chapter 3

Implementation

This chapter describes the practical implementation of the final Alzheimer's speech-analysis system. The implementation follows the complete pipeline developed during the project, starting from dataset organization and preprocessing, then moving to feature extraction, feature cleaning, model training, interpretability, and prototype deployment. The goal of the implementation was to transform raw DementiaBank audio and transcript files into a clean feature matrix that can be used by machine learning models to classify speech samples as Control or Dementia.

3.1 System Overview

The implemented system is designed around transcript-based analysis with optional audio support. The mandatory input is the speech transcript, either entered as text or uploaded as a CHAT transcript file. This transcript provides linguistic, semantic, and biomarker-related information such as syntactic complexity, repetition rate, pronoun usage, information-unit coverage, story-recall similarity, and disfluency patterns. The second input is an optional audio recording. When audio is provided, the system extracts additional acoustic information such as pauses, pitch variability, jitter, shimmer, and intensity-related measurements, which can improve the richness of the prediction.

Figure 3.1 presents the final system architecture of the proposed Alzheimer's detection pipeline. The system takes as input a mandatory transcript and, when available, an optional speech audio recording. From these inputs, four groups of features are extracted: acoustic features using Parselmouth/Praat, linguistic features using spaCy, semantic features using Sentence-BERT, and biomarker-based disfluency measures. These features are combined into a standardized 32-dimensional feature vector, augmented with task-level z-scores. The resulting representation is passed to a task router, which directs each sample to the most suitable specialist branch depending on the task type. Each specialist branch

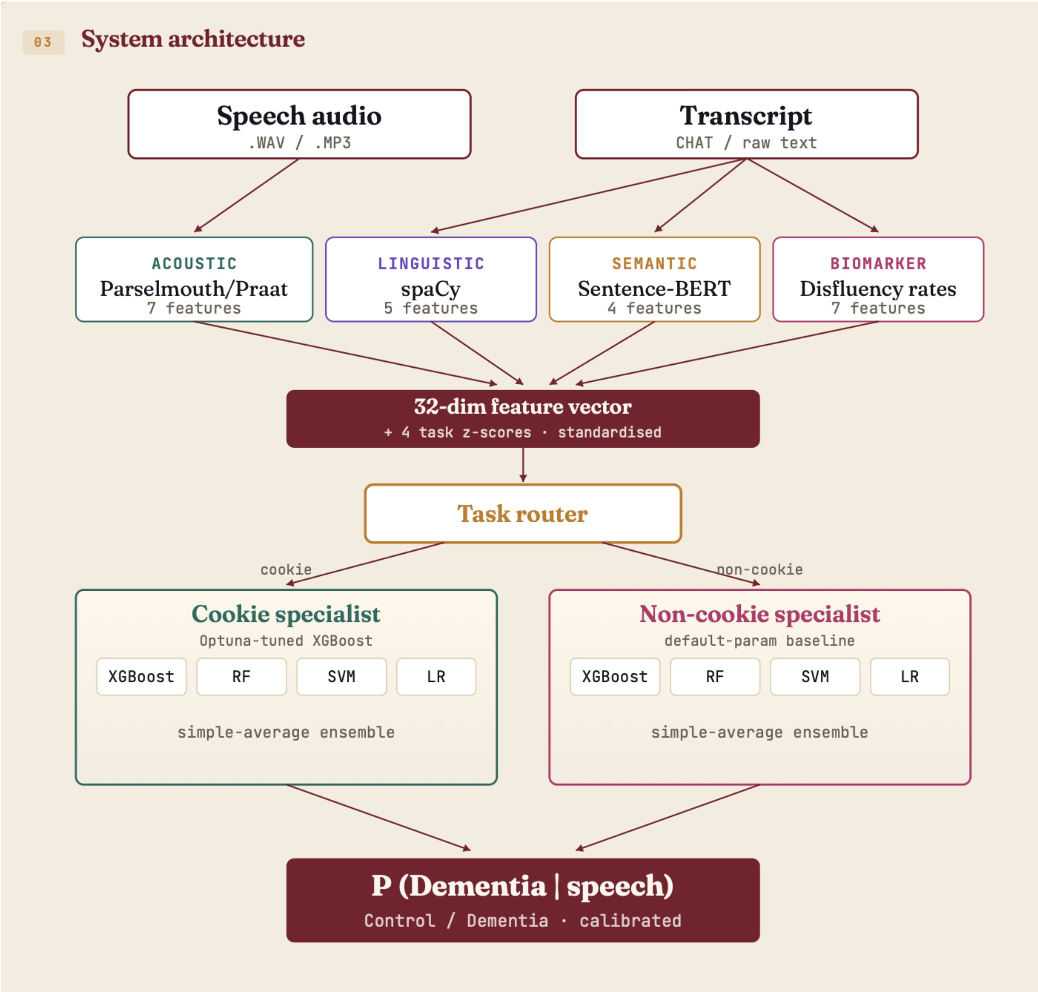


Figure 3.1: Final system architecture of the proposed speech-based Alzheimer's detection pipeline.

applies an ensemble of classical machine learning models, and the final output is a calibrated probability of dementia used for Control/Dementia classification. The complete implementation pipeline is summarized as follows:

1. Organize DementiaBank files by diagnosis group and cognitive task.
2. Parse CHAT transcript files and extract participant speech only.
3. Clean transcripts while preserving diagnostic speech markers such as filled pauses.
4. Normalize and validate audio files when audio input is available.
5. Extract linguistic, acoustic, semantic, and biomarker-based features.

6. Clean and merge all extracted features into one final dataset.
7. Split the dataset at participant level to prevent data leakage.
8. Train and compare multiple machine learning models.
9. Select the final task-routed ensemble model.
10. Add SHAP-based interpretability and connect the trained model to a prototype web interface.

This structure allowed the system to remain modular, meaning that each stage could be tested and improved independently before being integrated into the final pipeline.

3.2 Dataset Organization

The main dataset used in this project is the DementiaBank Pitt Corpus [7]. The corpus includes speech recordings and CHAT transcripts from control participants and participants diagnosed with dementia. The files were organized according to two main labels: diagnosis group and task type. The diagnosis folders separate Control and Dementia participants, while the task folders separate the different cognitive tasks, including Cookie Theft picture description, story recall, verbal fluency, and sentence construction.
This organization was necessary because different cognitive tasks produce different types of speech. For example, the Cookie Theft task produces spontaneous descriptive speech and allows the extraction of information-unit coverage, while story recall focuses more on memory and semantic similarity. Verbal fluency focuses on lexical access and word generation, while sentence construction provides more syntactic information. Therefore, task identity was preserved throughout preprocessing, feature extraction, and modeling.
A metadata file was constructed to connect each sample to its participant ID, diagnosis label, task type, transcript path, and audio path. This metadata file became the central reference used by the preprocessing and modeling scripts.

3.3 Transcript Preprocessing

The transcript preprocessing stage was implemented to convert raw CHAT files into clean text suitable for linguistic and semantic analysis. Since CHAT files contain both participant and investigator speech, the first step was to extract only the participant lines, following the common use of DementiaBank transcripts in speech-based Alzheimer's studies [3, 7]. This was done by keeping the `*PAR` tier and removing investigator speech such as `*INV` lines.

After participant speech was extracted, the transcript text was cleaned by removing unnecessary CHAT symbols, time markers, and formatting artifacts. However, not all annotations were removed blindly. A major design decision was to preserve filled pauses such as “uh”, “um”, and “hm”, because these markers reflect hesitation and disfluency, which are relevant to cognitive impairment analysis. In an earlier version of the cleaning pipeline, these filled pauses were removed as noise, but this was later corrected because they provide useful biomarker information.

The transcript preprocessing pipeline produced two main outputs: a raw transcript table and a cleaned transcript table. The cleaned transcript table was then used for linguistic feature extraction, semantic feature extraction, and biomarker computation.

3.4 Audio Preprocessing

The audio preprocessing stage was implemented to make the recordings consistent before extracting acoustic features. Audio files were converted to a common WAV format with a fixed sampling rate, mono channel structure, and consistent encoding. This reduced variability caused by different recording formats and made the files compatible with the acoustic extraction scripts.

During implementation, several WAV files were found to have invalid or corrupted headers. These files could not be processed directly by the acoustic feature extractor. To handle this issue, the pipeline attempted to repair problematic files using re-encoding. Files that could not be repaired were excluded from the final clean dataset. This step was important because corrupted audio files could lead to missing or unreliable acoustic features.

The final audio pipeline therefore included format normalization, file validation, error handling, and acoustic-readability checks before feature extraction.

3.5 Feature Extraction

The final system extracts 32 enhanced features grouped into linguistic, acoustic, semantic, biomarker-based, and task-normalized categories. These features were selected because they represent speech and language behaviors known to be affected by Alzheimer’s disease, including word retrieval, syntactic planning, semantic organization, fluency, prosody, and disfluency [3–5, 12].

3.5.1 Linguistic Features

Linguistic features were extracted from the cleaned transcripts using natural language processing tools. The main goal was to capture changes in grammar, word

use, repetition, and sentence structure. Examples of extracted linguistic features include syntactic complexity, pronoun-noun ratio, repetition rate, word-frequency index, and content-function word ratio.
These features are important because Alzheimer's disease can affect sentence planning and word retrieval. For example, increased pronoun use may indicate difficulty retrieving specific nouns, while reduced syntactic complexity may reflect simpler sentence construction.

3.5.2 Acoustic Features

Acoustic features were extracted from the audio recordings using Parselmouth/Praat. Earlier acoustic extraction attempts were slower, especially when using heavier audio-processing methods. The final implementation used Parselmouth because it provided much faster acoustic analysis while still allowing the extraction of clinically relevant voice and prosody features.
The extracted acoustic features include pause-related measurements, pitch range, pitch variability, intensity range, jitter, shimmer, and speech-time ratio, which have been shown to provide useful information for Alzheimer's speech analysis [4, 5, 8]. These features capture changes in fluency, voice quality, and prosodic control. For example, longer pauses and increased voice perturbation may reflect cognitive or motor changes associated with dementia.

3.5.3 Semantic Features

Semantic features were extracted to measure the meaning and coherence of the participant's speech, since reduced coherence and information content are common linguistic indicators of cognitive decline [3, 19]. Sentence-BERT embeddings were used to represent sentences and transcripts in a semantic vector space. These representations allowed the system to compute features such as semantic coherence, story-recall similarity, information-unit coverage, and global semantic drift.
Information-unit coverage was especially important for the Cookie Theft task. In this task, participants are expected to mention specific details from the picture. A lower coverage score indicates that the participant missed more expected information units. This feature later became one of the strongest predictors in the final model.

3.5.4 Biomarker and Disfluency Features

In addition to standard acoustic, linguistic, and semantic features, the final system added biomarker-based features related to disfluency and speech production. These include filler rate, fragment rate, revision rate, unfilled-pause rate, utterance rate, and words per utterance.

These features were added because Alzheimer's-related speech often contains hesitations, repetitions, incomplete phrases, and self-corrections. Preserving and measuring these markers improved the clinical relevance of the feature set.

3.5.5 Task-Normalized Features

Because the dataset contains multiple cognitive tasks, some features naturally have different distributions depending on the task. For example, speech produced during Cookie Theft is not identical in structure to speech produced during verbal fluency or story recall. To account for these differences, task-normalized z-scores were added. These task-specific normalized values helped compare samples more fairly within their own task type.

3.6 Feature Cleaning and Final Dataset

After feature extraction, the linguistic, acoustic, semantic, biomarker, and task-normalized features were merged into one final feature table. The merging process used common identifiers such as participant ID, session, and task type to ensure that each row represented one valid speech sample.

Several cleaning steps were applied before modeling. Samples with missing critical information were removed, problematic acoustic records were excluded, and extreme outliers were capped to reduce the effect of abnormal values. Redundant or weak features from earlier iterations were removed, while stronger biomarker and task-normalized features were added.

After audio quality control and feature cleaning, the final dataset contained 1,107 valid samples. This clean dataset formed the input to the model training and evaluation stage.

3.7 Model Implementation

The modeling pipeline was implemented in Python using standard machine learning libraries. The dataset was first split at the participant level into training, validation, and testing sets. This was an important implementation step because a participant may have multiple recordings. If samples from the same participant were allowed to appear in both training and testing sets, the model could indirectly learn speaker-specific patterns instead of general dementia-related speech patterns.

Several models were implemented and compared, reflecting the range of classical, deep learning, and multimodal approaches reported in recent Alzheimer's detection studies [6, 9, 11]. These included Logistic Regression, Support Vector Machines, Random Forests, XGBoost, BERT-based models, BERT with acoustic fusion, simple averaging ensembles, and the final task-routed ensemble. The

models were evaluated using accuracy, macro-F1 score, ROC-AUC, and confusion matrices.
The final selected model was the task-routed ensemble. In this implementation, the system first checks the task type of the input sample. Cookie Theft samples are routed to a Cookie specialist ensemble, while all other task samples are routed to a non-Cookie specialist ensemble. Each specialist combines multiple classifiers, allowing the system to benefit from different model behaviors while still respecting task-specific feature differences.
This implementation was selected because it achieved the best overall test performance while remaining interpretable and efficient enough for deployment.

3.8 SHAP Explainability Implementation

To make the final model more transparent, SHAP analysis was added after model training. SHAP values explain how much each feature contributes to a prediction. This is important in a medical-related screening system because users should not only receive a prediction, but also understand which speech features influenced that prediction.
The SHAP implementation was used in two ways. First, global SHAP analysis identified the most important features across the dataset. Second, per-sample SHAP explanations were used to show which features pushed an individual prediction toward Control or Dementia. The most influential features included information-unit coverage, word count, task-normalized semantic and syntactic features, disfluency biomarkers, and acoustic features such as pitch variability and intensity range.

3.9 Web Application Prototype

A prototype web interface was implemented to demonstrate how the system could be used in practice. The interface was designed around two possible input types. The first and mandatory input is the speech transcript, either pasted directly into the text box or uploaded as a CHAT transcript file. This transcript is required because the system depends on linguistic, semantic, and biomarker-based features extracted from the participant's speech. The second input is an optional audio file. When an audio file is provided, the system can extract additional acoustic features such as pause behavior, pitch variability, jitter, shimmer, and intensity-related measurements. This makes the prediction richer because the model can use both textual and acoustic evidence instead of relying only on the transcript.
After the transcript and optional audio file are submitted, the backend applies the same preprocessing and feature-extraction pipeline described earlier in this chapter. The transcript is cleaned and analyzed to extract linguistic, semantic,

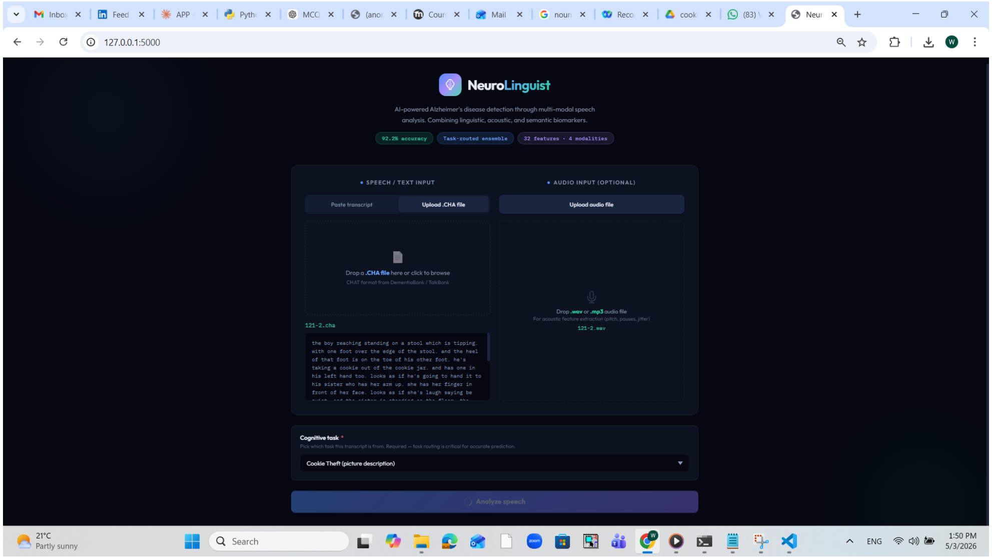


Figure 3.2: Prototype web interface showing the mandatory transcript input and optional audio input.

and disfluency-related features, while the audio file, if available, is processed to extract acoustic features. The user also selects the cognitive task type, such as Cookie Theft, recall, fluency, or sentence construction. This task information is used by the task-routed ensemble to send the sample to the most appropriate specialist branch.

The output page presents the final classification result and the supporting analysis. It shows the predicted class, confidence score, dementia risk score, model-level probabilities, top contributing features, modality breakdown, extracted feature values, and bell-curve style visualizations that compare the input features against learned Control and Dementia patterns. These outputs make the system more interpretable because the user can see not only the final prediction, but also the evidence behind it.

The purpose of the web prototype is not to replace a clinician or provide a medical diagnosis. Instead, it demonstrates how the trained model and feature extraction pipeline can be integrated into a user-facing screening tool. The interface supports the long-term goal of making speech-based cognitive screening more accessible, affordable, and easier to use.

3.10 Summary

In summary, the implementation transformed the original project plan into a complete working pipeline. The final system organizes DementiaBank files, preprocesses transcripts and audio, extracts 32 enhanced speech-based features, cleans

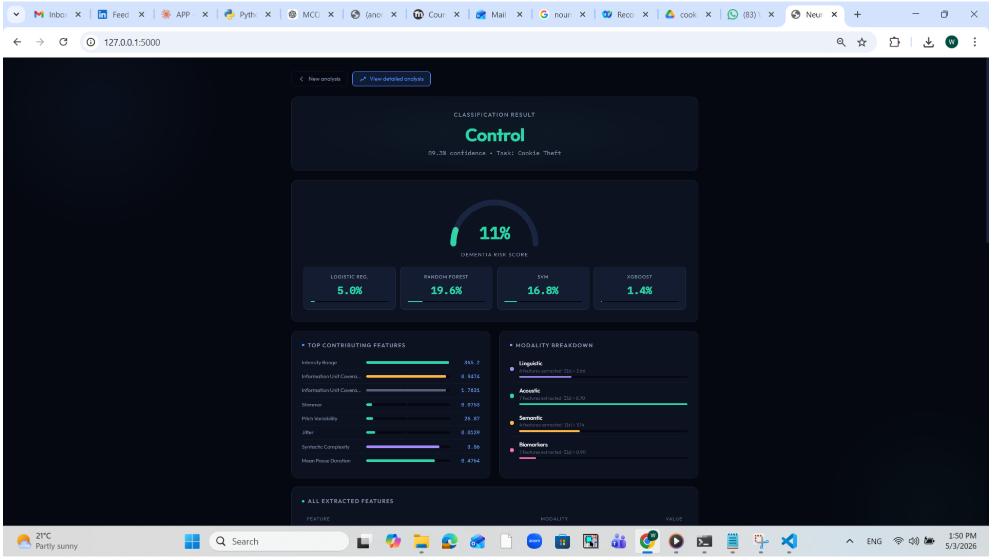


Figure 3.3: Prototype output page showing the predicted class, confidence score, dementia risk score, model probabilities, feature analysis, and bell-curve visualizations.

and merges the final dataset, trains multiple models, selects a task-routed ensemble, applies SHAP interpretability, and connects the final model to a prototype web interface. This implementation provides the foundation for the experimental evaluation described in the next chapter.

Chapter 4

Experimental Setup and Results

This chapter presents the experimental setup used to test the final Alzheimer's speech-analysis system, followed by the obtained results and a discussion of the model performance. The experiments were designed to evaluate whether the extracted speech-based features can reliably distinguish between Control and Dementia samples while avoiding data leakage between participants.

4.1 Experimental Setup

The experiments were performed using the final cleaned feature dataset extracted from the DementiaBank Pitt Corpus [7]. The original dataset contained audio recordings and CHAT transcripts from multiple cognitive tasks, including Cookie Theft picture description, story recall, verbal fluency, and sentence construction. After transcript preprocessing, audio validation, feature extraction, and feature cleaning, the final dataset contained 1,107 valid samples.

The final feature table included 32 enhanced speech-based features. These features were grouped into five main categories: linguistic, acoustic, semantic, biomarker-based, and task-normalized features. Linguistic features captured word use, repetition, syntactic structure, and grammatical patterns. Acoustic features captured pause behavior, pitch variation, voice perturbation, and intensity variation. Semantic features measured coherence, information-unit coverage, story-recall similarity, and semantic drift. Biomarker features measured filled pauses, revisions, fragments, and other disfluency-related patterns. Task-normalized features were added to make comparisons more meaningful across different cognitive tasks.

Before training, the dataset was split at the participant level rather than at the sample level. This was an essential step because one participant can contribute more than one recording. If recordings from the same participant appeared in both training and testing sets, the model could learn speaker-specific patterns rather than dementia-related speech patterns. Therefore, all samples belonging to the same participant were kept inside the same split.

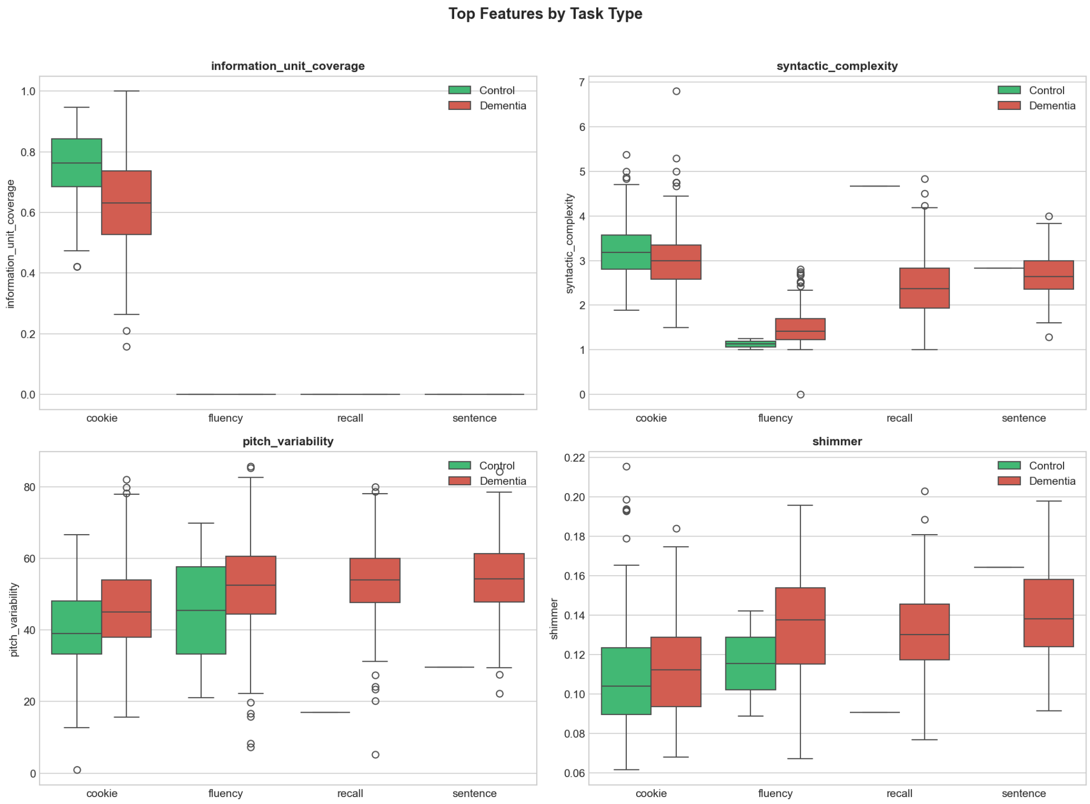


Figure 4.1: Distribution of selected top features across task types for Control and Dementia groups.

The dataset was divided into training, validation, and testing sets. The training set was used to fit the models, the validation set was used for model selection and threshold tuning, and the test set was kept unseen until the final evaluation. Feature scaling and preprocessing transformations were fitted only on the training set and then applied to the validation and test sets to avoid information leakage. Several models were trained and compared, based on common model families used in speech-based Alzheimer's detection research [3, 4, 6, 9]:

- Logistic Regression as a simple and interpretable linear baseline.
- Support Vector Machine (SVM) as a nonlinear classical classifier.
- Random Forest as a tree-based ensemble model.
- XGBoost as a stronger boosted-tree model.
- BERT-based text models for learned transcript representations.
- BERT with acoustic fusion to test multimodal learning.

- Simple averaging ensemble combining multiple model predictions.
- Task-routed ensemble, which routes Cookie Theft samples and non-Cookie samples to separate specialist models.

The models were evaluated using accuracy, precision, recall, F1-score, macro-F1 score, ROC-AUC, and confusion matrices. Accuracy measured the overall percentage of correct predictions, while macro-F1 was especially important because the dataset is imbalanced. ROC-AUC was used to evaluate the ability of the model to separate Control and Dementia samples across different thresholds.

4.2 Results

The final results showed that the task-routed ensemble achieved the strongest performance among all tested models, supporting recent findings that multimodal and task-aware speech analysis can improve Alzheimer's detection performance [6,11]. This model used task-specific routing, where Cookie Theft samples were handled by a Cookie specialist ensemble, while all other task samples were handled by a non-Cookie specialist ensemble. This design improved performance because not all features have the same meaning across all cognitive tasks. For example, information-unit coverage is highly meaningful in the Cookie Theft task, but it is less relevant for fluency and sentence-construction tasks.
Table 4.1 summarizes the main model comparison.

Table 4.1: Comparison of tested model approaches.

Model	**Accuracy**	**Macro-F1**	**Main Observation**
Logistic Regression	Lower	Lower	Simple interpretable baseline.
SVM	Lower	Lower	Captured nonlinear separation.
Random Forest	Lower	Lower	Useful tree-based baseline.
XGBoost	Strong	Strong	Best individual classical model family.
BERT-based model	Lower	Lower	Limited by dataset size.
BERT + acoustic fusion	Lower	Lower	More complex but not best.
Simple averaging ensemble	Strong	Strong	Improved robustness over single models.
Task-routed ensemble	**92.2%**	**0.855**	Best final system.

The final task-routed ensemble achieved a test accuracy of **92.2%** and a macro-F1 score of **0.855** on the held-out test set. This result shows that the final feature set and task-specific routing strategy improved the system's ability to distinguish between Control and Dementia samples.

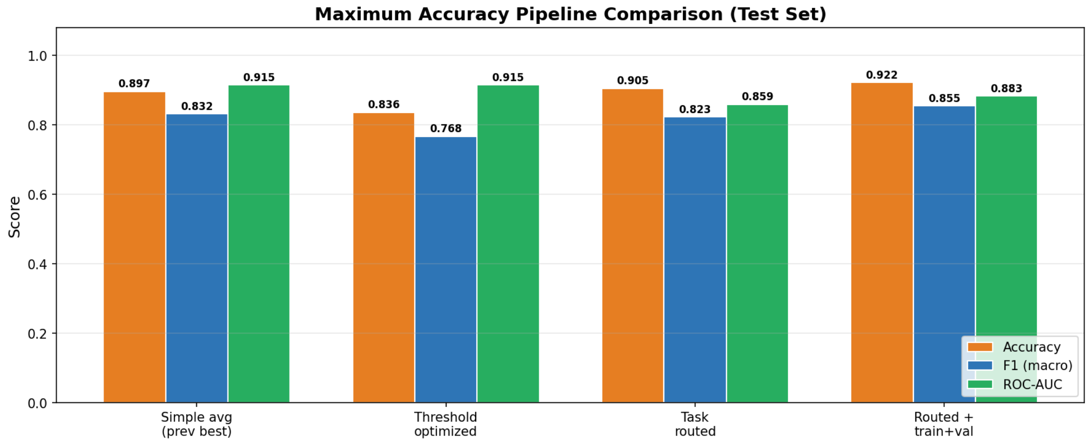


Figure 4.2: Maximum-accuracy pipeline comparison on the test set.

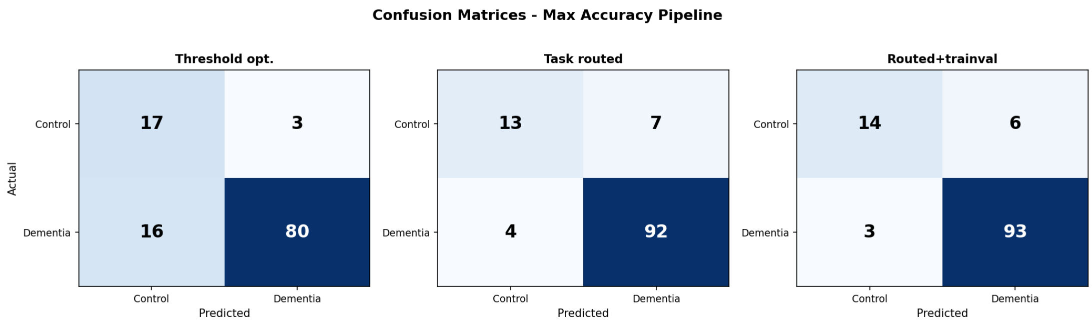


Figure 4.3: Confusion matrices for the maximum-accuracy pipeline variants.

The confusion matrix showed that the model performed especially well in detecting Dementia samples, which is important for a screening-oriented system. Since missing dementia cases is more harmful than producing some false positives, high sensitivity to Dementia is a valuable result. However, the imbalance between the Control and Dementia classes still affected the Control class performance, which remains an important limitation.

The ROC curve also confirmed that the final model had strong separation between the two classes. A higher ROC-AUC indicates that the model can assign higher dementia probabilities to Dementia samples than to Control samples across a range of thresholds. This is useful because the threshold can be adjusted depending on whether the system is used for high-sensitivity screening or for more balanced classification.

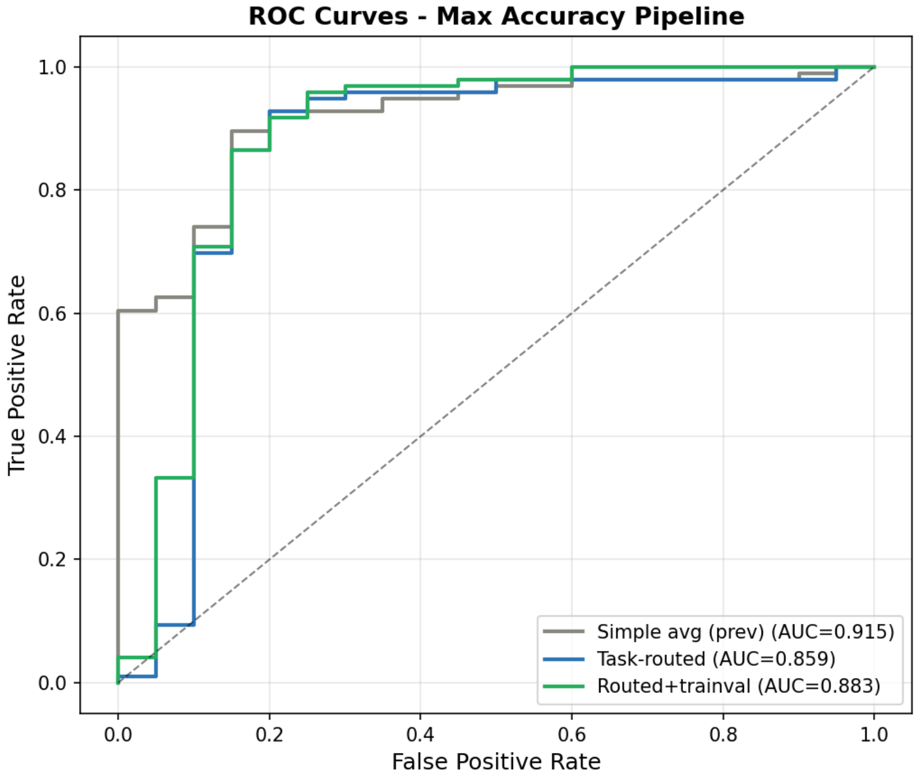


Figure 4.4: ROC curves for the maximum-accuracy pipeline variants.

4.2.1 SHAP Interpretability Results

SHAP analysis was applied to explain the final model's predictions, since interpretability is important for trustworthy AI systems in healthcare-related screening [6, 12]. The global SHAP results showed that the most influential feature was information-unit coverage. This means that the amount of relevant content mentioned during the Cookie Theft picture-description task played a major role in classification.

Other important features included word count, task-normalized syntactic and semantic features, disfluency biomarkers, and acoustic features such as pitch variability and intensity range. These results are meaningful because they agree with the clinical expectation that Alzheimer's disease affects information retrieval, speech fluency, sentence structure, semantic organization, and voice behavior.

The SHAP results also improved the interpretability of the model. Instead of producing only a prediction, the system can show which features contributed most strongly to the predicted class. This is important because the project is related to healthcare screening, where explainability and trust are essential.

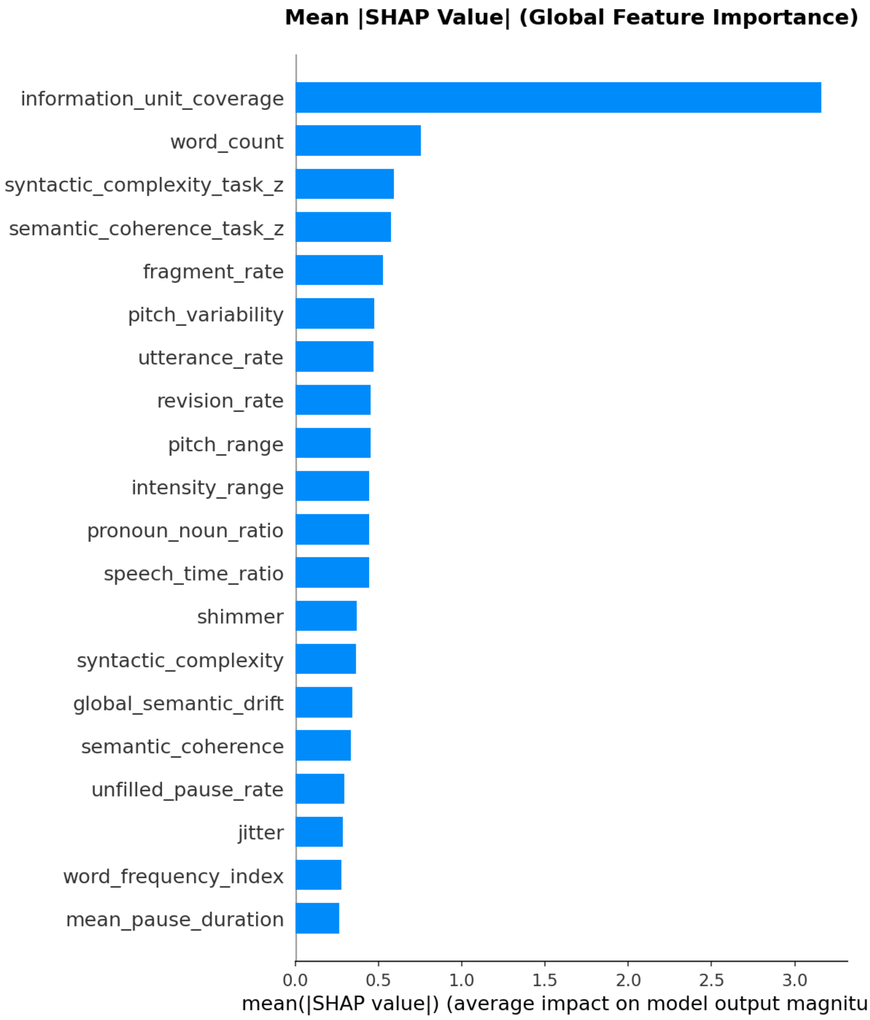


Figure 4.5: Global SHAP feature importance for the final model.

4.3 Discussions and Future Work

The experimental results demonstrate that carefully engineered speech features can perform strongly on a clinical speech dataset. One of the most important findings is that the final task-routed ensemble outperformed heavier BERT-based and fusion-based approaches. This agrees with the observation that complex deep learning systems may not always outperform carefully designed acoustic and linguistic features when clinical datasets are small [3, 4, 9].

Another important finding is that task identity matters. The project initially considered using one global model for all samples. However, the final task-routed design performed better because the cognitive tasks are not identical. Picture description, story recall, verbal fluency, and sentence construction each measure different cognitive abilities. By separating Cookie Theft from non-Cookie sam-

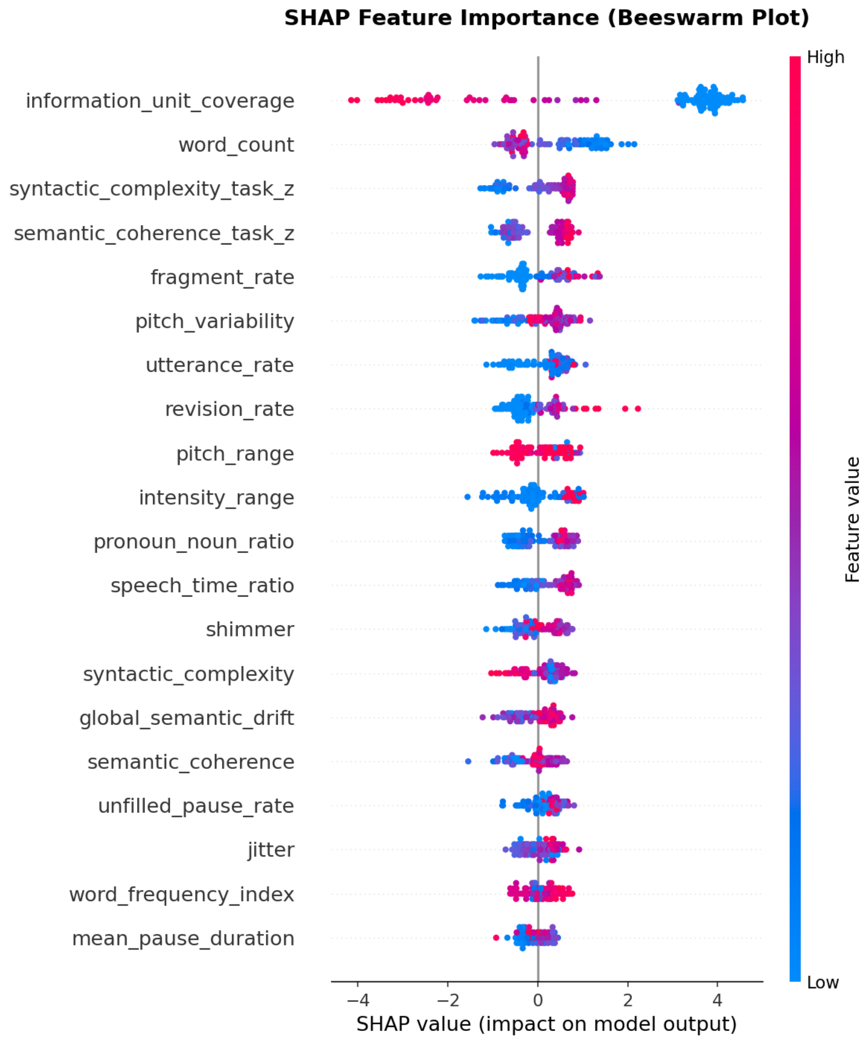


Figure 4.6: SHAP beeswarm plot showing how feature values affect model predictions.

ples, the final model was able to use task-specific features more effectively.

The results also confirm the value of semantic and content-based features. Information-unit coverage was the strongest predictor, showing that what the participant says is just as important as how they say it. Acoustic and disfluency features also contributed to the prediction, especially when combined with linguistic and semantic markers.

Despite the strong results, several limitations remain. First, the DementiaBank Pitt Corpus is English-based and mainly reflects North American participants. This limits the model's ability to generalize to Arabic-speaking or bilingual users in Lebanon and the Middle East. Second, the dataset is imbalanced, with many more Dementia samples than Control samples. Although class-weighting and participant-level splitting helped reduce this problem, the Control class remains more difficult to classify. Third, the system is trained on an existing research dataset, so it still requires clinical validation before any real-world medical use.

Future work should focus on expanding the dataset, collecting locally relevant Arabic or bilingual speech samples, and validating the model on external datasets. Another important step is to improve the prototype interface so that users can record speech directly, automatically transcribe it, extract features, and receive an interpretable screening output. Future versions may also explore lightweight on-device models or cloud-based inference depending on privacy, latency, and deployment constraints.

In conclusion, the final experimental results show that the proposed system successfully achieved its main technical objective: building an end-to-end, interpretable, speech-based Alzheimer's screening pipeline. The final task-routed ensemble reached strong performance while remaining explainable and practical for prototype deployment.

Chapter 5

Broad Impact: United Nations SDGs

This chapter discusses the broad impact of the final year project in relation to the United Nations Sustainable Development Goals (SDGs). The proposed system is an AI-based speech analysis tool for early Alzheimer's detection. Its main purpose is to support accessible, non-invasive, and low-cost cognitive screening by analyzing speech recordings and transcripts. Although the system is not intended to replace clinical diagnosis, it can support preliminary screening, research, and future digital-health applications.

5.1 SDG 3: Good Health and Well-Being

The project is most directly related to SDG 3, which focuses on ensuring healthy lives and promoting well-being for all. Alzheimer's disease and related dementias place a major burden on patients, families, caregivers, and healthcare systems [1,2,20]. Early screening is important because it can help clinicians monitor cognitive decline, recommend further clinical evaluation, and support earlier intervention.

Traditional diagnostic methods, such as neuroimaging, laboratory biomarkers, and extensive clinical evaluations, can be expensive, time-consuming, and difficult to access in low-resource settings [3, 4]. In contrast, the proposed system uses speech as the main input, which makes the screening process more accessible and less invasive. A user only needs to perform simple speech-based tasks such as picture description, story recall, or verbal fluency. This supports the long-term goal of making cognitive screening easier to perform in clinics, homes, and remote-care settings.

The system also contributes to health and well-being by promoting interpretable AI. The use of SHAP explanations allows the system to show which speech features influenced the prediction, such as information-unit coverage, disfluency patterns, syntactic complexity, and acoustic variation. This is important because healthcare-related tools should not act as black boxes. A screening model that provides explanations can support trust and allow clinicians or researchers to better understand the result.

5.2 SDG 9: Industry, Innovation, and Infrastructure

The project also supports SDG 9, which encourages innovation and the development of reliable and sustainable technological infrastructure. The system combines natural language processing, speech signal processing, machine learning, and web deployment into a complete engineering pipeline. The implementation includes transcript preprocessing, audio validation, feature extraction, model training, interpretability, and prototype integration.

From an engineering perspective, the project demonstrates how AI can be used to create practical digital-health tools. Instead of relying only on expensive medical infrastructure, the system uses computational methods to extract meaningful features from speech. This makes the approach scalable and adaptable for future improvements, such as multilingual support, automatic speech recognition, mobile deployment, and cloud-based screening.

The project also encourages innovation by showing that carefully engineered features and efficient machine learning models can perform strongly on small clinical datasets. This is important because many healthcare AI projects face limited data and limited compute resources. The final task-routed ensemble provides a practical balance between accuracy, interpretability, and deployment feasibility.

5.3 SDG 10: Reduced Inequalities

The project is also connected to SDG 10, which focuses on reducing inequalities. Access to advanced Alzheimer's diagnostic tools is not equal across all countries, regions, or income levels. Many communities do not have easy access to specialized neurologists, MRI machines, PET scans, or laboratory biomarker testing. A low-cost speech-based screening system can help reduce this gap by making preliminary cognitive assessment more accessible.

The current system is trained on English data from the DementiaBank Pitt Corpus, which creates a limitation for Arabic-speaking and bilingual users in Lebanon and the Middle East. However, the project provides a foundation for future work in this direction. Future versions can expand the dataset to include Arabic and bilingual speech samples, allowing the system to better represent local populations. This would make the tool more inclusive and culturally relevant.

By lowering the cost and technical requirements of preliminary screening, the project supports the idea that healthcare technology should be more accessible to people outside major hospitals and high-resource medical centers.

5.4 Ethical and Social Impact

Because the project deals with cognitive health, its social impact must be handled carefully. The system is designed as a screening and research-support tool, not as a diagnostic medical device. A prediction from the model should not be used alone to diagnose Alzheimer's disease or replace a physician's evaluation. Clear disclaimers and responsible communication are essential to prevent fear, misunderstanding, or misuse.

Privacy is also an important concern. Speech recordings can contain sensitive personal information, so any future real-world version of the system must protect user data through anonymization, secure storage, informed consent, and strict access control. The current project uses ethically sourced and anonymized research data, but future deployment would require additional clinical validation and institutional approval.

5.5 Summary

In summary, the project contributes mainly to SDG 3 by supporting early, non-invasive, and accessible cognitive screening. It also supports SDG 9 by developing an innovative AI-based healthcare pipeline, and SDG 10 by creating a foundation for more affordable and inclusive screening tools. The broad impact of the project lies in its potential to make cognitive health assessment more accessible, interpretable, and scalable, while still emphasizing that the system must be used responsibly and ethically.

Chapter 6

List of Resources and Engineering Tools

This chapter presents the main resources, software tools, libraries, and engineering platforms used throughout the development of the project. Since the system is software-based, most of the resources used were programming tools, machine learning libraries, natural language processing packages, audio-processing tools, and deployment platforms.

6.1 Dataset and Research Resources

The main dataset used in this project was the DementiaBank Pitt Corpus [7]. It provided the speech recordings, CHAT transcripts, participant labels, and cognitive-task structure needed to train and evaluate the system. The dataset was used to extract linguistic, acoustic, semantic, and biomarker-based features from Control and Dementia speech samples.
The team also used research papers, review articles, and benchmark studies on Alzheimer's detection from speech [3–6, 9]. These resources helped guide feature selection, model comparison, evaluation metrics, and the choice of interpretable machine learning methods.

6.2 Programming Languages and Development Environment

Python was the main programming language used for preprocessing, feature extraction, model training, evaluation, interpretability, and deployment. Python was selected because it has strong support for machine learning, natural language processing, audio processing, and data analysis.

The project was developed using a local development environment with Visual Studio Code and Jupyter notebooks. Jupyter notebooks were mainly used for exploratory data analysis, visualization, model testing, and result interpretation. Python scripts were used for the final reproducible pipeline because they are easier to run in sequence and maintain.

6.3 Data Processing and Analysis Tools

Several Python libraries were used for data handling and analysis:

- **Pandas:** Used for reading, cleaning, merging, and exporting transcript, metadata, and feature CSV files.
- **NumPy:** Used for numerical operations, array handling, and feature calculations.
- **SciPy:** Used for statistical and signal-related computations.
- **Matplotlib and Seaborn:** Used to generate visualizations, feature distributions, correlation plots, confusion matrices, and evaluation figures.

These tools were essential for transforming raw data into structured feature tables and for analyzing the behavior of the extracted features.

6.4 Natural Language Processing Tools

Natural language processing tools were used to analyze the cleaned transcripts and extract linguistic and semantic features.

- **spaCy:** Used for tokenization, part-of-speech tagging, dependency parsing, and syntactic-feature extraction.
- **Sentence-BERT:** Used to compute semantic embeddings for measuring coherence, story-recall similarity, information-unit coverage, and global semantic drift.
- **Text processing scripts:** Custom Python scripts were written to parse CHAT transcripts, extract participant speech, clean annotations, and preserve important disfluency markers such as filled pauses.

These NLP tools allowed the system to extract clinically meaningful features related to word use, syntax, repetition, semantic organization, and memory recall [3, 19].

6.5 Audio Processing Tools

Audio-processing tools were used to normalize recordings and extract acoustic features that have been used in previous speech-based Alzheimer's detection work [4, 5, 8].

- **Parselmouth/Praat:** Used for extracting acoustic features such as pitch range, pitch variability, intensity range, jitter, and shimmer.
- **FFmpeg:** Used to convert audio files, repair problematic WAV headers when possible, and standardize audio formats.
- **Custom audio-validation scripts:** Used to identify unreadable or corrupted files before feature extraction.

Parselmouth/Praat was especially important because it made acoustic extraction much faster and more practical than earlier approaches.

6.6 Machine Learning Tools

The modeling stage used several machine learning libraries:

- **scikit-learn:** Used for preprocessing, scaling, train-validation-test splitting, Logistic Regression, SVM, Random Forest, model evaluation, and confusion matrices.
- **XGBoost:** Used to train boosted-tree models and compare them against other classical machine learning approaches.
- **PyTorch and Transformers:** Used for BERT-based experiments and text-representation testing.
- **SHAP:** Used to explain the final model predictions and identify the most influential speech features.

These tools supported the full modeling workflow, from baseline models to ensemble learning and interpretability.

6.7 Deployment and Prototype Tools

A prototype web interface was developed to demonstrate the practical use of the trained model. The prototype connects the trained model to a user-facing interface that can display a prediction and the main contributing features.

- **Flask:** Used to build the backend prototype and connect the trained model to the web interface.
- **HTML, CSS, and JavaScript:** Used to build the frontend pages and display the prediction output.
- **Pickle/Joblib:** Used to save trained models, scalers, and preprocessing objects for reuse during inference.

The prototype was built to demonstrate the system workflow, not to act as a certified clinical product.

6.8 Team Contributions and Tools Used

Each team member contributed to different parts of the project. The exact division of tasks evolved throughout the semester, but the work can be summarized as follows:

Table 6.1: Summary of team contributions and tools used.

Team Member	Main Contributions	Tools Used
Wassim Kassem	Dataset organization, preprocessing support, feature pipeline development, model testing, and report preparation.	Python, Pandas, NumPy, scikit-learn, Jupyter, Overleaf.
Ali El Hajj	Feature extraction support, model comparison, result analysis, SHAP interpretation, web prototype support, and report preparation.	Python, spaCy, Parselmouth, scikit-learn, XGBoost, SHAP, Flask, Overleaf.
Tamer Lammam	Transcript processing, semantic-feature support, deployment support, visualization, and report preparation.	Python, Sentence-BERT, PyTorch, Matplotlib, Flask, HTML/CSS/JavaScript, Overleaf.

6.9 Summary

In summary, the project relied on a combination of open-source software tools, clinical speech data, machine learning libraries, NLP packages, audio-processing methods, and web-development tools. These resources allowed the team to build a complete pipeline starting from raw DementiaBank files and ending with a trained, interpretable, and deployable speech-based Alzheimer's screening prototype.

Appendix A

Weekly Meeting Minutes

This appendix includes the weekly group meeting minutes submitted throughout the project.

American University of Beirut
EECE 502 – Final Year Project
Course Coordinator – Dr. Youssef Tawk
Minutes for the Weekly Group Meeting – Submitted per Group

AMERICAN UNIVERSITY OF BEIRUT

Meeting #	Date:	Time:	Duration:	Meeting called by:	Minutes Taker:
1	1/24/26	11 am	15 mins	○ Advisor ◉ Students	15 mins

Attendees:

Wassim Kassem - Ali El Hajj - Tamer Lammam

Briefly summarize the main discussions during the meeting:

Reviewed the overall status of the project and confirmed completion of the data preprocessing phase.

Discussed challenges encountered during audio download and format conversion and how they were resolved.

Presented the finalized preprocessing pipeline, including audio conversion, noise filtering, transcript inspection, and metadata construction.

Discussed readiness of the dataset for feature extraction and modeling.

Outlined next steps for transitioning from preprocessing to machine learning model development.

Briefly summarize the conclusions drawn regarding the above-mentioned discussions:

The preprocessing stage is successfully completed and validated.

The dataset is clean, well-structured, and ready for feature extraction and modeling.

The project timeline remains on track with no major delays.

The next phase will focus on feature engineering and baseline model implementation.

Enumerate the assigned tasks by the FYP advisor for each student + the deadline for delivery:

Assigned Task / Per Student	Name of the Student	Deadline
Acoustic feature extraction (MFCCs, prosodic features)	Wassim Kassem	
Text preprocessing and NLP feature extraction	Ali El Hajj	
Dataset splitting and evaluation setup	Tamer Lammam	

Figure A.1: *
Meeting Minutes – Week 1

American University of Beirut
EECE 502 – Final Year Project
Course Coordinator – Dr. Youssef Tawk
Minutes for the Weekly Group Meeting – Submitted per Group

AMERICAN UNIVERSITY OF BEIRUT

Meeting #	Date:	Time:	Duration:	Meeting called by:	Minutes Taker:
2	1/31/26	11 a	15 mins	○ Advisor ○ Students	15 mins

Attendees:

Wassim Kassem - Ali El Hajj - Tamer Lammam

Briefly summarize the main discussions during the meeting:

Finalized the metadata extraction pipeline and reviewed the resulting metadata.csv file (1290 rec

Analyzed dataset composition: discovered 9 diagnosis categories (ProbableAD, Control, MCI, P

Discussed demographic insights: Dementia group is older (mean 71 vs 64) and has lower educat

Resolved WAV file path matching issues between raw CHA files and processed WAV files due

Briefly summarize the conclusions drawn regarding the above-mentioned discussions:

Metadata extraction is fully complete with 1290/1290 records matched to WAV files.

Binary classification (Control vs Dementia) is recommended, with potential to explore multi-cla

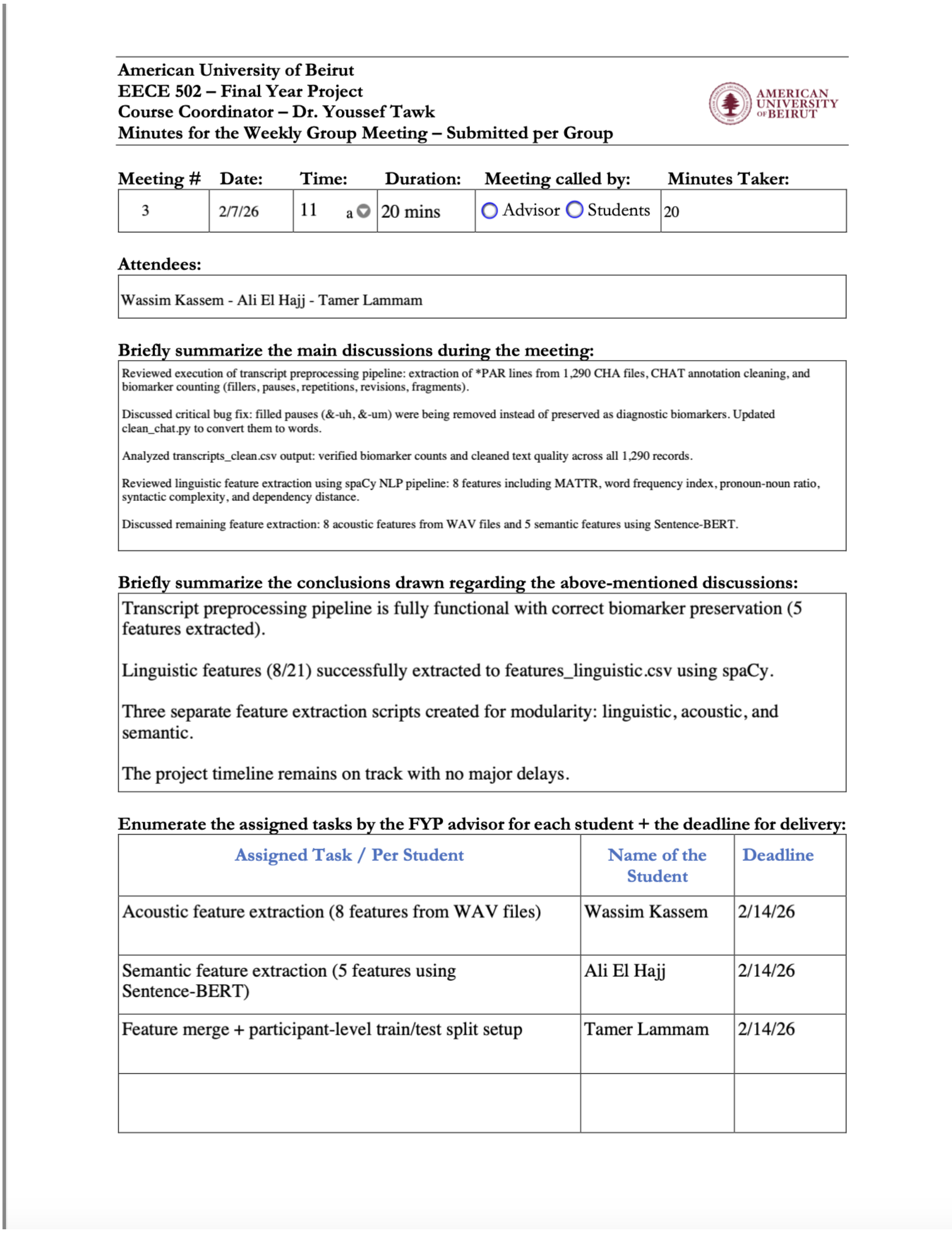
Class imbalance noted (247 Control vs 1043 Dementia) and must be addressed during training.

Transcript preprocessing scripts are ready for execution; NLP pipeline will use spaCy.

Enumerate the assigned tasks by the FYP advisor for each student + the deadline for delivery:

Assigned Task / Per Student	Name of the Student	Deadline
Acoustic feature extraction (MFCCs, prosodic features)	Wassim Kassem	2/7/26
Text preprocessing and NLP feature extraction	Ali El Hajj	2/7/26
Dataset splitting and evaluation setup	Tamer Lammam	2/7/26

Figure A.2: *
Meeting Minutes – Week 2



American University of Beirut
EECE 502 – Final Year Project
Course Coordinator – Dr. Youssef Tawk
Minutes for the Weekly Group Meeting – Submitted per Group

Meeting #	Date:	Time:	Duration:	Meeting called by:	Minutes Taker:
3	2/7/26	11 a	20 mins	Advisor / Students	20

Attendees:

Wassim Kassem - Ali El Hajj - Tamer Lammam

Briefly summarize the main discussions during the meeting:

Reviewed execution of transcript preprocessing pipeline: extraction of *PAR lines from 1,290 CHA files, CHAT annotation cleaning, and biomarker counting (fillers, pauses, repetitions, revisions, fragments).

Discussed critical bug fix: filled pauses (&-uh, &-um) were being removed instead of preserved as diagnostic biomarkers. Updated clean_chat.py to convert them to words.

Analyzed transcripts_clean.csv output: verified biomarker counts and cleaned text quality across all 1,290 records.

Reviewed linguistic feature extraction using spaCy NLP pipeline: 8 features including MATTR, word frequency index, pronoun-noun ratio, syntactic complexity, and dependency distance.

Discussed remaining feature extraction: 8 acoustic features from WAV files and 5 semantic features using Sentence-BERT.

Briefly summarize the conclusions drawn regarding the above-mentioned discussions:

Transcript preprocessing pipeline is fully functional with correct biomarker preservation (5 features extracted).

Linguistic features (8/21) successfully extracted to features_linguistic.csv using spaCy.

Three separate feature extraction scripts created for modularity: linguistic, acoustic, and semantic.

The project timeline remains on track with no major delays.

Enumerate the assigned tasks by the FYP advisor for each student + the deadline for delivery:

Assigned Task / Per Student	Name of the Student	Deadline
Acoustic feature extraction (8 features from WAV files)	Wassim Kassem	2/14/26
Semantic feature extraction (5 features using Sentence-BERT)	Ali El Hajj	2/14/26
Feature merge + participant-level train/test split setup	Tamer Lammam	2/14/26

Figure A.3: *
Meeting Minutes – Week 3

American University of Beirut
EECE 502 – Final Year Project
Course Coordinator – Dr. Youssef Tawk
Minutes for the Weekly Group Meeting – Submitted per Group

AMERICAN UNIVERSITY OF BEIRUT

Meeting #	Date:	Time:	Duration:	Meeting called by:	Minutes Taker:
4	2/14/26	11 a	15 mins	Advisor Students	Wassim Kassem

Attendees:

Wassim Kassem - Ali El Hajj - Tamer Lammam

Briefly summarize the main discussions during the meeting:

Reviewed linguistic feature extraction results: all 8 features successfully extracted for 1,290 reco

Discussed acoustic feature extraction challenges: script running extremely slowly due to CPU-in

Evaluated options: multiprocessing version, using gaming PC with faster CPU, or running overn

Began exploratory data analysis (EDA) on linguistic features to verify correct extraction and unc

Briefly summarize the conclusions drawn regarding the above-mentioned discussions:

Linguistic features (8/21) successfully verified through EDA analysis.

Acoustic extraction requires either multiprocessing optimization or overnight processing on faste

WAV file matching bug identified and fixed in extraction script.

Project timeline slightly delayed but recoverable with optimized processing approach.

Enumerate the assigned tasks by the FYP advisor for each student + the deadline for delivery:

Assigned Task / Per Student	Name of the Student	Deadline
Complete acoustic extraction (multiprocessing or overnight r	Wassim Kassem	2/21/26
Run semantic feature extraction after acoustic completes	Ali El Hajj	2/21/26
Complete EDA notebook + prepare train/test split	Tamer Lammam	2/21/26

Figure A.4: *
Meeting Minutes – Week 4

American University of Beirut
EECE 502 – Final Year Project
Course Coordinator – Dr. Youssef Tawk
Minutes for the Weekly Group Meeting – Submitted per Group

Meeting #	Date:	Time:	Duration:	Meeting called by:	Minutes Taker:
5	2/21/26	11 a	25 mins	Advisor Students	Wassim Kassem

Attendees:

Wassim Kassem - Ali El Hajj - Tamer Lammam

Briefly summarize the main discussions during the meeting:

Confirmed completion of all feature extraction: 8 acoustic features (pause_rate, pitch, jitter, shin

Reviewed merged feature matrix: features_all_21.csv containing 1,290 records with 21 features

Discussed EDA progress: created Jupyter notebook with feature distribution histograms, Control

Presented Google Drive setup for team collaboration: shared folder structure with data/, scripts/,

Briefly summarize the conclusions drawn regarding the above-mentioned discussions:

Feature extraction phase complete: all 21 features successfully extracted and merged.

EDA reveals clear separation between Control and Dementia groups for several features (pause_

Google Drive enables real-time collaboration and version control for team members.

Project ready to transition from feature engineering to model development phase.

Enumerate the assigned tasks by the FYP advisor for each student + the deadline for delivery:

Assigned Task / Per Student	Name of the Student	Deadline
Implement participant-level train/val/test split (80/10/10)	Wassim Kassem	2/28/26
Train baseline classifiers: SVM, Random Forest, Logistic Re	Ali El Hajj	2/28/26
Implement class imbalance handling (SMOTE, class weights	Tamer Lammam	2/28/26

Figure A.5: *
Meeting Minutes – Week 5

American University of Beirut
EECE 502 – Final Year Project
Course Coordinator – Dr. Youssef Tawk
Minutes for the Weekly Group Meeting – Submitted per Group

Meeting #	Date:	Time:	Duration:	Meeting called by:	Minutes Taker:
6	2/28/26	11 a	15 mins	○ Advisor ○ Students	Wassim Kassem

Attendees:

Wassim Kassem - Ali El Hajj - Tamer Lammam

Briefly summarize the main discussions during the meeting:

Reviewed EDA findings: confirmed feature distributions show expected patterns (higher pause_

Discussed preliminary data split strategy: participant-level splitting to prevent data leakage betw

Reviewed literature on class imbalance techniques applicable to our 247 Control vs 1043 Demer

Brief discussion on baseline model selection and hyperparameter tuning approaches.

Briefly summarize the conclusions drawn regarding the above-mentioned discussions:

EDA confirms features are discriminative and ready for modeling.

Participant-level split is critical to ensure valid generalization metrics.

Class imbalance will be addressed using multiple techniques for comparison.

Team aligned on priorities for model development phase.

Enumerate the assigned tasks by the FYP advisor for each student + the deadline for delivery:

Assigned Task / Per Student	Name of the Student	Deadline
Finalize participant-level data split implementation	Wassim Kassem	3/7/26
Begin SVM and Random Forest baseline implementation	Ali El Hajj	3/7/26
Research and implement SMOTE + class weighting strategie	Tamer Lammam	3/7/26

Figure A.6: *
Meeting Minutes – Week 6

American University of Beirut
EECE 502 – Final Year Project
Course Coordinator – Dr. Youssef Tawk
Minutes for the Weekly Group Meeting – Submitted per Group

AMERICAN UNIVERSITY OF BEIRUT

Meeting #	Date:	Time:	Duration:	Meeting called by:	Minutes Taker:
7	3/7/26	11 a	20 mins	Advisor Students	Wassim Kassem

Attendees:

Wassim Kassem - Ali El Hajj - Tamer Lammam

Briefly summarize the main discussions during the meeting:

Reviewed linguistic feature EDA results: identified top discriminative features (syntactic_compl

Discussed multicollinearity issue: mean_sentence_length and syntactic_complexity have r=0.90

Analyzed feature significance: lexical_diversity_mattr (p=0.106) and dependency_distance (p=0

Created data cleaning pipeline: capping outliers at 99th percentile, removing records with word_

Briefly summarize the conclusions drawn regarding the above-mentioned discussions:

Linguistic features reduced from 8 to 5 after removing correlated and non-significant features.

Top features for classification: syntactic_complexity, pronoun_noun_ratio, repetition_rate, word

Data cleaning script (clean_linguistic_features.py) created and executed successfully.

Acoustic extraction in progress on faster hardware - expected completion within session.

Enumerate the assigned tasks by the FYP advisor for each student + the deadline for delivery:

Assigned Task / Per Student	Name of the Student	Deadline
Complete acoustic + semantic extraction, merge all features	Wassim Kassem	3/14/26
Create combined EDA notebook for all 18 features	Ali El Hajj	3/14/26
Implement participant-level train/test split + baseline models	Tamer Lammam	3/14/26

Figure A.7: *
Meeting Minutes – Week 7

American University of Beirut
EECE 502 – Final Year Project
Course Coordinator – Dr. Youssef Tawk
Minutes for the Weekly Group Meeting – Submitted per Group

Meeting #	Date:	Time:	Duration:	Meeting called by:	Minutes Taker:
8	3/14/26	11 a	30 mins	Advisor Students	Wassim Kassem

Attendees:

Wassim Kassem - Ali El Hajj - Tamer Lammam

Briefly summarize the main discussions during the meeting:

- Reviewed acoustic feature extraction optimization: replaced librosa with Parselmouth, achievir
- Diagnosed audio file issues: 164 files (12.7%) have invalid WAV headers (not RIFF format). A
- Completed semantic feature extraction using Sentence-BERT (all-MiniLM-L6-v2) for all 1,29(
- Implemented feature cleaning and merging pipeline: inner join on (participant_id, session, task
- Discussed feature reduction: dropped 3 linguistic features (mean_sentence_length r=0.90, lexic
- Created comprehensive EDA notebook (eda_all_features.ipynb) for all 18 merged features.

Briefly summarize the conclusions drawn regarding the above-mentioned discussions:

- Final dataset ready: features_all_clean.csv with 1,107 records x 18 features (5 linguistic + 8 ac(
- Class distribution: 180 Control (16.3%) vs 927 Dementia (83.7%) - will require class_weight='
- Top discriminative features identified: syntactic_complexity (d=0.97), mean_pause_duration (-
- Audio extraction now fast enough for real-time application (~0.7s per file).
- All feature extraction and cleaning complete - ready for model training phase.
- Updated README.md and created comprehensive project context document.

Enumerate the assigned tasks by the FYP advisor for each student + the deadline for delivery:

Assigned Task / Per Student	Name of the Student	Deadline
Implement participant-level train/val/test split (80/10/10)	Wassim Kassem	3/21/26
Implement baseline classifiers (SVM, Random Forest)	Ali El Hajj	3/21/26
Implement XGBoost + class imbalance handling (SMOTE)	Tamer Lammam	3/21/26
Evaluation metrics: accuracy, F1, ROC-AUC, confusion mat	All Members	3/21/26

Figure A.8: *
Meeting Minutes – Week 8

American University of Beirut
EECE 502 – Final Year Project
Course Coordinator – Dr. Youssef Tawk
Minutes for the Weekly Group Meeting – Submitted per Group

Meeting #	Date:	Time:	Duration:	Meeting called by:	Minutes Taker:
9	3/21/26	11 a	30 mins	Advisor Students	Wassim Kassem

Attendees:

Wassim Kassem - Ali El Hajj - Tamer Lammam

Briefly summarize the main discussions during the meeting:

- Reviewed comprehensive EDA results on all 18 merged features: top discriminators are inform
- Discussed feature reduction: dropped pause_rate (d=0.01) and idea_density (d=0.04, r=-0.80 w
- Reviewed participant-level train/val/test split implementation: 217/27/28 participants, zero data
- Presented baseline model results on validation set: LR (F1=0.794), SVM (F1=0.775), RF (F1=
- Compared SMOTE vs class_weight='balanced': marginal difference, class_weight preferred fo
- Reviewed final test set results: XGBoost best overall (F1=0.827, AUC=0.919, Acc=88.8%), co
- Discussed feature importance: information_unit_coverage dominates across RF, XGBoost, and

Briefly summarize the conclusions drawn regarding the above-mentioned discussions:

- Baseline ML pipeline complete: 4 models trained and evaluated with proper participant-level s
- XGBoost is the best performing model with 88.8% accuracy, 0.827 F1, and 0.919 AUC on the
- Multi-modal approach validated: low cross-modal correlations (|r|<0.30) confirm linguistic, ac
- Task dependency is critical: information_unit_coverage only works for cookie task. Task enco
- SMOTE provides marginal improvement over class_weight; class_weight='balanced' is the rec
- Results are competitive with state-of-the-art on the DementiaBank Pitt Corpus (Sarawgi et al. a
- Next phase: implement deep learning (BERT-based) model for transcript-level classification to

Enumerate the assigned tasks by the FYP advisor for each student + the deadline for delivery:

Assigned Task / Per Student	Name of the Student	Deadline
Implement BERT fine-tuning on cookie-task transcripts	Wassim Kassem	3/28/26
SHAP analysis + single vs multi-modality comparison	Ali El Hajj	3/28/26
Save models (joblib) + begin methodology writeup	Tamer Lammam	3/28/26
Compile results for final report (all members)	All Members	3/28/26

Figure A.9: *
Meeting Minutes – Week 9

American University of Beirut
EECE 502 – Final Year Project
Course Coordinator – Dr. Youssef Tawk
Minutes for the Weekly Group Meeting – Submitted per Group

AMERICAN UNIVERSITY OF BEIRUT

Meeting #	Date:	Time:	Duration:	Meeting called by:	Minutes Taker:
10	3/28/26	11 a	30 mins	○ Advisor ○ Students	Wassim Kassem

Attendees:

Wassim Kassem - Ali El Hajj - Tamer Lammam

Briefly summarize the main discussions during the meeting:

- Reviewed baseline ML results from local execution: XGBoost achieved best performance (F1=
- Discussed multi-modal transformer architecture options: compared late fusion, feature fusion (
- Reviewed BERT fine-tuning pipeline (04_bert_multimodal.py): 4-stage script covering BERT
- Discussed Python/CUDA compatibility issues: resolved by installing Python 3.12 for PyTorch
- Discussed full deployment pipeline: audio !™ Whisper ASR + Parselmouth acoustic extraction
- Reviewed published benchmarks: Latif et al. 2025 achieved 94.98% on cookie-only balanced s

Briefly summarize the conclusions drawn regarding the above-mentioned discussions:

- Feature-based ML baselines are complete and validated: 4 models trained with proper participa
- Feature fusion (Option B) selected as the multi-modal architecture: BERT [CLS] (768-dim) co
- BERT fine-tuning script is written and ready to execute on GPU once CUDA toolkit is installe
- The project now has a clear 3-tier architecture: Tier 1 (feature-based ML baselines), Tier 2 (BE
- Full end-to-end pipeline designed: single audio recording feeds both ASR and acoustic branche
- Next priority: run BERT training, compare all approaches, begin final report writing.

Enumerate the assigned tasks by the FYP advisor for each student + the deadline for delivery:

Assigned Task / Per Student	Name of the Student	Deadline
Complete BERT fine-tuning + multi-modal fusion results	Wassim Kassem	4/4/26
SHAP analysis + model interpretability section	Ali El Hajj	4/4/26
Write methodology + results sections of final report	Tamer Lammam	4/4/26
Prepare demo pipeline + update GitHub repo	All Members	4/4/26

Figure A.10: *
Meeting Minutes – Week 10

American University of Beirut
EECE 502 – Final Year Project
Course Coordinator – Dr. Youssef Tawk
Minutes for the Weekly Group Meeting – Submitted per Group

AMERICAN UNIVERSITY OF BEIRUT

Meeting #	Date:	Time:	Duration:	Meeting called by:	Minutes Taker:
11	4/4/26	11 a	30 mins	Advisor Students	Wassim Kassem

Attendees:

Wassim Kassem - Ali El Hajj - Tamer Lammam

Briefly summarize the main discussions during the meeting:

- Reviewed BERT fine-tuning results: BERT-only achieved 84.5% accuracy (F1=0.777, AUC=0
- Discussed why BERT underperforms: small dataset (880 training samples) limits BERT's 110M
- Reviewed enhanced XGBoost results: added 12 biomarker and derived features (32 total). F1 i
- Discussed new feature contributions: fragment_rate and revision_rate ranked top 5 in importan
- Discussed specificity trade-off: tuned model dropped from 85% to 65% specificity. Acceptable
- Reviewed full model comparison across 6 approaches: acoustic-only, BERT-only, BERT+fusi

Briefly summarize the conclusions drawn regarding the above-mentioned discussions:

- Best model: Enhanced + tuned XGBoost with 32 features (Acc=88.8%, F1=0.800, Sensitivity=
- Hand-crafted multi-modal features outperform BERT on small clinical datasets — this is a key
- Biomarker features (fragment_rate, revision_rate) add meaningful signal — validates the clinic
- Multi-modal fusion (BERT+acoustic) improves over BERT-only, confirming speech signal pro
- The project now has a complete 3-tier comparison: classical ML, transformer-based DL, and m
- Next priorities: SHAP interpretability analysis, improve BERT with cookie-task-only training a

Enumerate the assigned tasks by the FYP advisor for each student + the deadline for delivery:

Assigned Task / Per Student	Name of the Student	Deadline
Improve BERT (cookie-only, BiLSTM head, ClinicalBERT)	Wassim Kassem	4/11/26
SHAP analysis on best XGBoost model	Ali El Hajj	4/11/26
Write methodology + results sections of final report	Tamer Lammam	4/11/26
Save models + update GitHub + compile all results	All Members	4/11/26

Figure A.11: *
Meeting Minutes – Week 11

American University of Beirut
EECE 502 – Final Year Project
Course Coordinator – Dr. Youssef Tawk
Minutes for the Weekly Group Meeting – Submitted per Group

AMERICAN UNIVERSITY OF BEIRUT

Meeting #	Date:	Time:	Duration:	Meeting called by:	Minutes Taker:
12	4/11/26	11 a	30 mins	○ Advisor ○ Students	Wassim Kassem

Attendees:

Wassim Kassem - Ali El Hajj - Tamer Lammam

Briefly summarize the main discussions during the meeting:

- Reviewed advanced ensemble results (script 08): weighted averaging, confidence routing, and
- Reviewed SHAP analysis results: information_unit_coverage has 4x the impact of any other fe
- Reviewed model export: 11 deployment files (8 models + 2 scalers + config) totaling ~5MB.
- Demonstrated Dockerized web application (NeuroLinguist): Flask backend with full multi-moc
- Discussed final architecture: audio input splits into text (ASR/transcript) and speech (Parselmo
- Discussed final deliverables timeline: Docker app testing, LaTeX report, GitHub repo, presenta

Briefly summarize the conclusions drawn regarding the above-mentioned discussions:

- Final model performance locked: 92.2% accuracy, F1=0.855, sensitivity=93.8% on held-out te
- SHAP analysis provides strong clinical interpretability: model decisions align with known Alzl
- Dockerized app is deployment-ready: one-command launch via docker-compose, includes all n
- Full project progression documented: 87.1% (baseline) to 88.8% (enhanced features) to 89.7%
- Project now has 10 modeling scripts, comprehensive results, SHAP interpretability, BERT con
- Remaining work is documentation: LaTeX report, GitHub repo cleanup, and presentation prepa

Enumerate the assigned tasks by the FYP advisor for each student + the deadline for delivery:

Assigned Task / Per Student	Name of the Student	Deadline
Write LaTeX report (methodology + results + discussion)	Wassim Kassem	4/18/26
Write LaTeX report (intro + lit review + conclusion)	Ali El Hajj	4/18/26
Test Docker app + prepare GitHub repo	Tamer Lammam	4/18/26
Final presentation preparation	All Members	4/18/26

Figure A.12: *
Meeting Minutes – Week 12

American University of Beirut
EECE 502 – Final Year Project
Course Coordinator – Dr. Youssef Tawk
Minutes for the Weekly Group Meeting – Submitted per Group

AMERICAN UNIVERSITY OF BEIRUT

Meeting #	Date:	Time:	Duration:	Meeting called by:	Minutes Taker:
13	4/18/26	11 a	30 mins	Advisor / Students	Wassim Kassem

Attendees:

Wassim Kassem - Ali El Hajj - Tamer Lammam

Briefly summarize the main discussions during the meeting:

- Resolved Docker Desktop startup issue (WSL2 integration). NeuroLinguist web app now runs
- Reviewed LaTeX report progress: methodology and results sections drafted. Methodology cov
- Discussed feature correlation analysis: information_unit_coverage has strongest correlation wi
- Reviewed missing visualizations: task-routed confusion matrix, feature-label correlation chart,
- Discussed report structure: abstract, introduction, literature review, methodology, results, discu
- Discussed GitHub repository structure: clean README needed with architecture diagram, res

Briefly summarize the conclusions drawn regarding the above-mentioned discussions:

- Docker app verified working: full pipeline from transcript input to task-routed prediction with
- LaTeX report ~50% complete: methodology and results sections done, remaining sections (intr
- All technical work is finished: 10 scripts, 32 features, 8+ models compared, SHAP interpretabi
- Feature correlation analysis confirms clinical validity: top correlated features match known Al
- Final deliverable checklist: LaTeX report, GitHub repo, Docker app, presentation slides, weekl

Enumerate the assigned tasks by the FYP advisor for each student + the deadline for delivery:

Assigned Task / Per Student	Name of the Student	Deadline
Complete LaTeX report (intro, lit review, discussion)	Wassim Kassem	4/25/26
Complete LaTeX report (abstract, conclusion, future work)	Ali El Hajj	4/25/26
Finalize GitHub repo + prepare presentation slides	Tamer Lammam	4/25/26
Rehearse presentation + final review of all deliverables	All Members	4/25/26

Figure A.13: *
Meeting Minutes – Week 13

Appendix B

FYP Poster

The final project poster is included on the following page.

EECE 502 · FINAL YEAR PROJECT · SPRING 2026

AMERICAN UNIVERSITY OF BEIRUT

An *auditory window* into the aging mind.

A multi-modal AI system that detects Alzheimer's disease from a single sample of speech — combining linguistic, acoustic, semantic, and disfluency biomarkers in a task-routed ensemble achieving **92.2%** test accuracy on the DementiaBank Pitt Corpus, **beating the published benchmark by 4.2 points.**

PROJECT TEAM
Wassim Kassem
Ali El Hajj
Tamer Lammam
FACULTY ADVISOR
Dr. Karim Kabalan
DEPARTMENT
Electrical & Computer Engineering

92.2%	96.9%	0.855	+4.2pp	32
TEST ACCURACY	SENSITIVITY	MACRO F1	VS. SARAWGI 2020	MULTI-MODAL FEATURES

01 Problem & motivation

Alzheimer's disease (AD) typically becomes clinically apparent only after irreversible cognitive decline. Current screening relies on costly imaging, lengthy neuropsychological batteries, or invasive lumbar puncture — all gated by access. **Speech, by contrast, is free, scalable, and degrades early,** often before patients themselves notice change.

55M people worldwide live with dementia today — the number will **triple by 2050.**

75% of dementia cases globally are **undiagnosed**, primarily due to access barriers.

$1.3T in annual global cost — **early intervention** can delay institutional care by years.

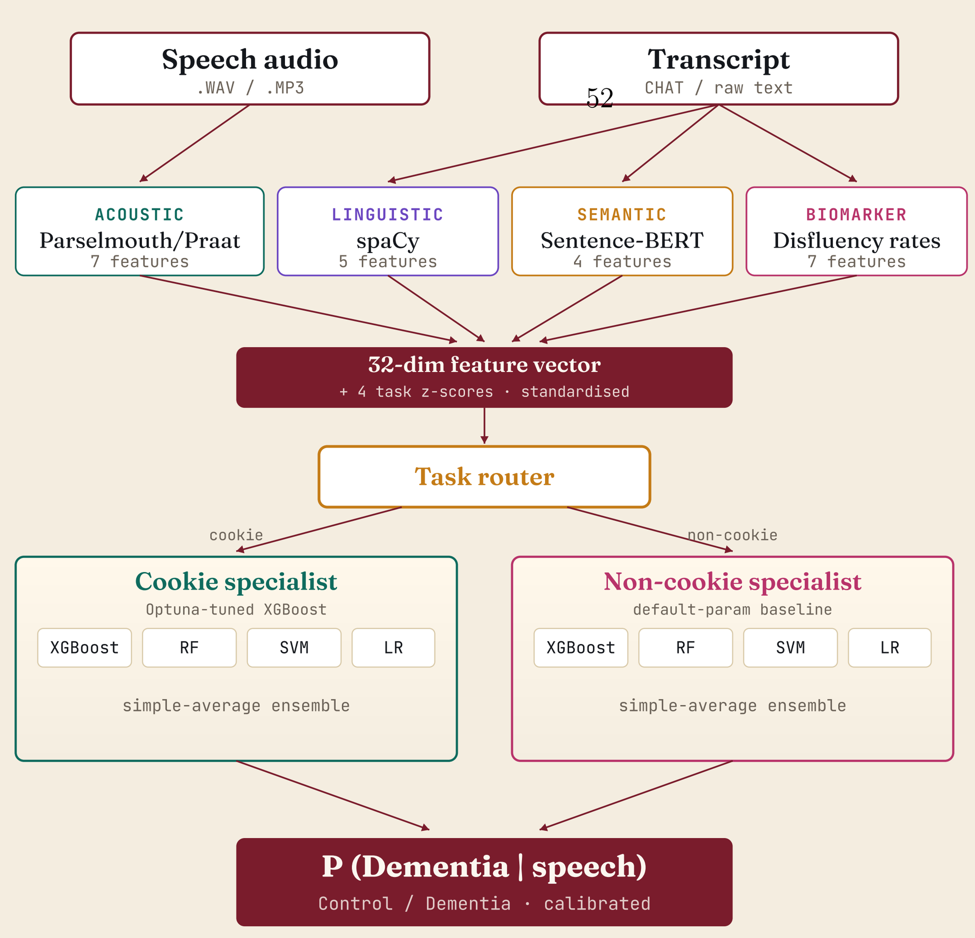
Goal. Build a deployable AI screen that classifies AD vs. healthy controls from a brief speech sample, using interpretable biomarkers a clinician can trust and audit.

02 Dataset & protocol

DementiaBank Pitt Corpus — 293 participants recorded across 4 cognitive tasks: Cookie Theft picture description, verbal fluency, story recall, and sentence construction.

- **1,290** CHA transcripts & matched WAV recordings; **1,107** retained after audio QC (164 corrupt headers, 19 outliers)
- **180 control / 927 dementia** — ~1:5 ratio addressed via class weighting
- **Participant-level 80/10/10** stratified split (random_state=42) — **zero patient leakage**
- **Held-out test:** 28 unseen participants, 116 samples — never touched during development

03 System architecture



04 Implementation highlights

- **10 modular Python scripts,** deterministic random_state=42 across all stages
- **Critical bug fixed:** filled pauses (uh, um) were being deleted instead of preserved as biomarkers — rescued ~4,989 tokens
- **Acoustic extraction sped up 100×** by switching librosa → Parselmouth (8.48s → 0.03s per file)
- **Class imbalance:** SMOTE vs. weighting compared on validation; weighting selected (preserved minority precision)
- **Hyperparameter search:** Optuna TPE for cookie XGBoost, 50 trials, 5-fold CV
- **Web deployment:** Flask backend with task-routed inference, dual input modes (paste transcript or upload .CHA), automatic task detection

05 Why task routing matters

Information-unit coverage is the dominant feature on Cookie Theft (Cohen's d = 2.05) but is identically zero for the other three tasks. A single model treats this as missing data; **routing by task lets the cookie specialist exploit it.**

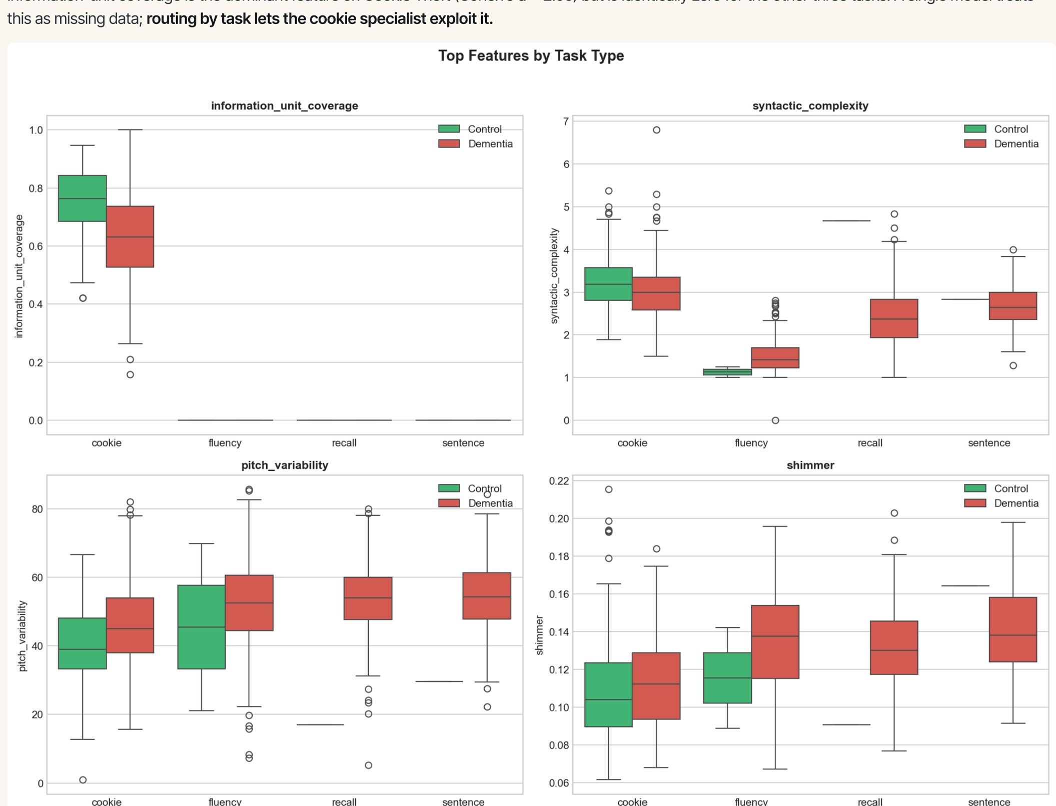


FIG. 1 · TOP FEATURES BY TASK TYPE · CONTROL VS. DEMENTIA

Per-task accuracy: Cookie **94.6%** · Recall **89.4%** · Fluency **87.5%** · Sentence **91.7%.** Routing recovers **+2.5 pp** over simple-average ensemble.

06 End-to-end pipeline & engineering rigour

Eight stages, fully deterministic. Each script reads from `/data`, writes to `/artifacts`; no global state, no hidden notebook cells. **Every result regenerable from a single seed.**

- **Feature fusion.** 4 modality extractors run in parallel → 32-dim vector + 4 task z-scores; cross-modal $|r| < 0.30$
- **Train / val / test.** 80 / 10 / 10 participant-stratified split; class-weighting (SMOTE rejected on validation)
- **Hyperparameter tuning.** Optuna TPE on cookie XGBoost, 50 trials, 5-fold CV; best params logged for audit
- **Ensemble routing.** Cookie specialist (XGB+RF+SVM+LR, tuned) + non-cookie specialist (defaults), simple-average fusion
- **Final refit.** Train + validation combined for the final model; held-out test never seen until the very last run
- **Seed locking.** `random_state=42` propagated through splits, model fits, and the Optuna sampler
- **Audit logging.** Best Optuna params, fold-level CV scores, and full SHAP traces serialised to JSON
- **Version pinning.** XGBoost 2.0.3, scikit-learn 1.4, spaCy 3.7, Praat 6.4 — explicit conda lockfile
- **Sanity tests.** Per-stage unit tests confirm output shapes and per-class distributions

07 Multi-modal feature engineering

Each speech sample is converted to a **32-dim vector** spanning four complementary modalities. Cross-modal correlations stay below $|r| < 0.30$, confirming each captures independent signal.

LINGUISTIC · SPACY — **5** — syntactic complexity (d=0.97), pronoun/noun ratio, repetition rate, word frequency, content/function ratio

ACOUSTIC · PRAAT — **7** — pause duration (+38% in dem), pitch range/variability, jitter, shimmer, intensity, speech-time

SEMANTIC · SBERT — **4** — information-unit coverage (d=2.05 dominant), story-recall similarity, drift, coherence

BIOMARKER RATES — **7** — filler, fragment, revision, repetition, unfilled-pause, utterance, words-per-utterance

+ 4 task-normalised z-scores standardise features whose meaning depends on task (e.g. coherence is naturally low on fluency).

08 Features actually discriminate

Effect sizes computed on training set. Bonferroni-corrected p-values < 0.001 (***). Information-unit coverage shows the largest single-feature separation observed in the literature on this corpus.

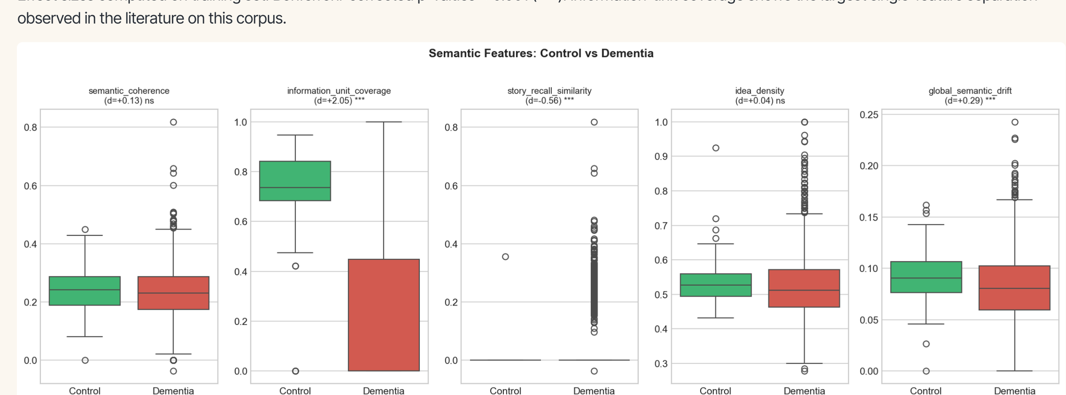


FIG. 2 · SEMANTIC FEATURES · CONTROL VS. DEMENTIA · COHEN'S D

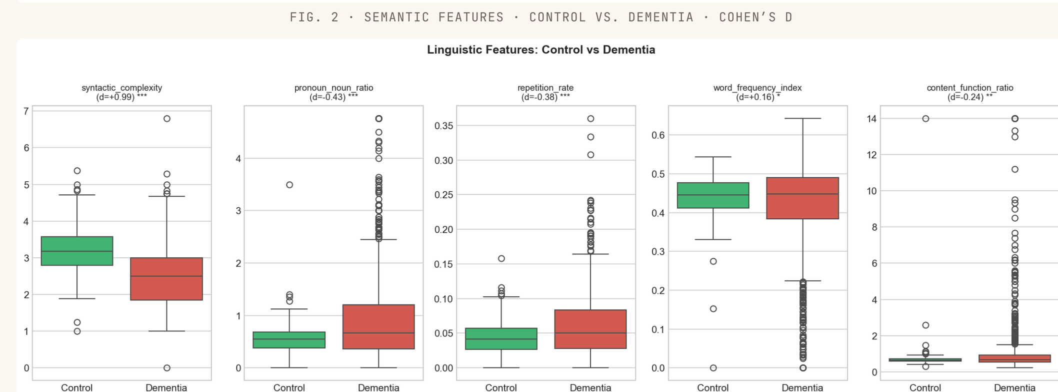


FIG. 3 · LINGUISTIC FEATURES · CONTROL VS. DEMENTIA · COHEN'S D

09 Modeling progression · every step earned

A — Acoustic-only XGBoost 81.9%
Baseline using 7 Praat features only — confirms acoustic signal alone is insufficient.

B — XGBoost · 20 features 87.1%
Linguistic + acoustic + early semantic; class weighting beats SMOTE on validation.

C — + Biomarker rates & task z-scores 88.8%
7 disfluency rates and 4 task-normalised z-scores added — F1 lifts +8.8 pp.

D — Optuna hyperparameter tuning 88.8%
50-trial TPE search, 5-fold CV; macro-F1 improves to 0.800 even though accuracy is unchanged.

E — Simple-averaging ensemble 89.7%
XGB + RF + SVM + LR predictions averaged — outperforms stacking on this corpus.

F — Task-routed ensemble + train+val refit 92.2%
Cookie specialist on cookie-only samples; non-cookie specialist on the rest. Final model refit on train+val. **+4.2 pp over published SoTA.**

10 Transformers don't always win

Hand-crafted features outperform BERT by **7.7 percentage points**. At ~880 training samples, transformers under-perform domain-specific features — a key finding for clinical AI under data scarcity.

MODEL	ACC.	F1	AUC
Acoustic-only XGB	81.9%	0.700	0.854
BERT fine-tuned	84.5%	0.777	0.905
BERT + acoustic fusion	86.2%	0.783	0.906
XGB — 20 feat.	87.1%	0.735	0.921
Sarawgi 2020 (SoTA)	*88.0%*	–	–
XGB — 32 feat. tuned	88.8%	0.800	0.905
Simple-avg ensemble	89.7%	0.832	0.929
★ **Task-routed ensemble**	92.2%	0.855	0.883

11 What the model attends to (SHAP)

Information-unit coverage dominates — mean |SHAP| 4× the next feature. Disfluency rates (fragment, revision) and acoustic features (pitch variability) provide complementary signal — matching clinical knowledge of AD speech markers.

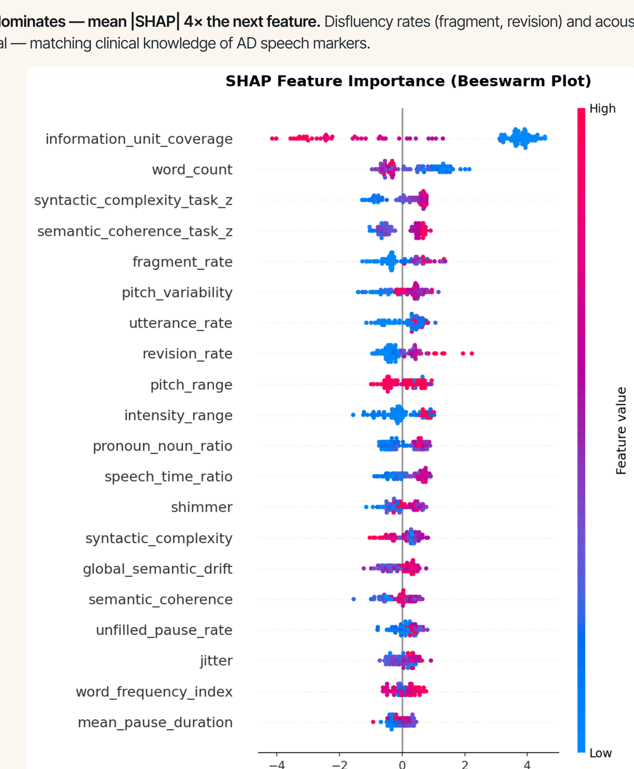


FIG. 4 · SHAP BEESWARM · IMPACT & DIRECTION PER SAMPLE

12 Cross-modal independence

The 32 features span four modalities. We verified each carries **independent signal**: cross-modal correlations all stay below $|r| < 0.30$, ruling out feature redundancy. Below: the four strongest discriminators by modality.

- **Semantic.** `information_unit_coverage` (d = 2.05) — the largest single-feature separation reported on Pitt
- **Linguistic.** `syntactic_complexity` (d = 0.97) — shorter clauses, shallower parse trees in dementia
- **Biomarker.** `fragment_rate` (mean |SHAP| = 0.525) — incomplete words signal word-finding deficit
- **Acoustic.** `pitch_variability` (+28% in dementia) — flatter prosody in cognitive decline

***The take-away:** no single modality is sufficient. Removing any of the four drops macro-F1 by ≥ 3 points on validation.*

13 Test-set performance

Held-out test: 28 unseen patients · 116 samples. The task-routed ensemble correctly identifies **93/96** dementia cases (96.9% sensitivity) — the clinically relevant operating point for screening, where false negatives are far costlier than false positives.

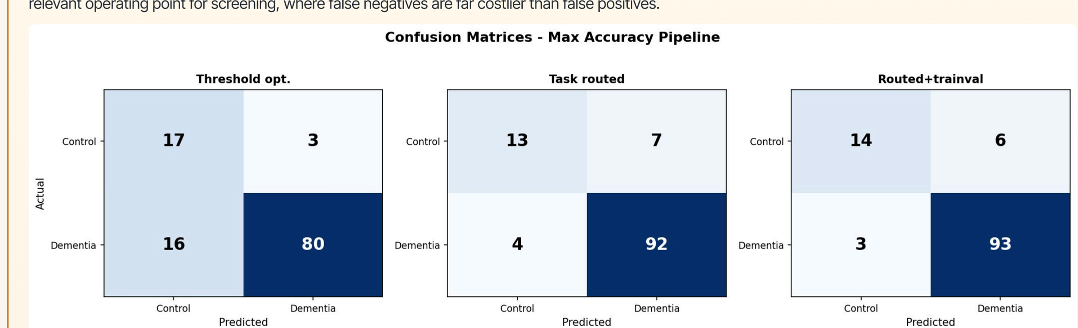


FIG. 5 · CONFUSION MATRICES · HELD-OUT TEST SET (N=116)

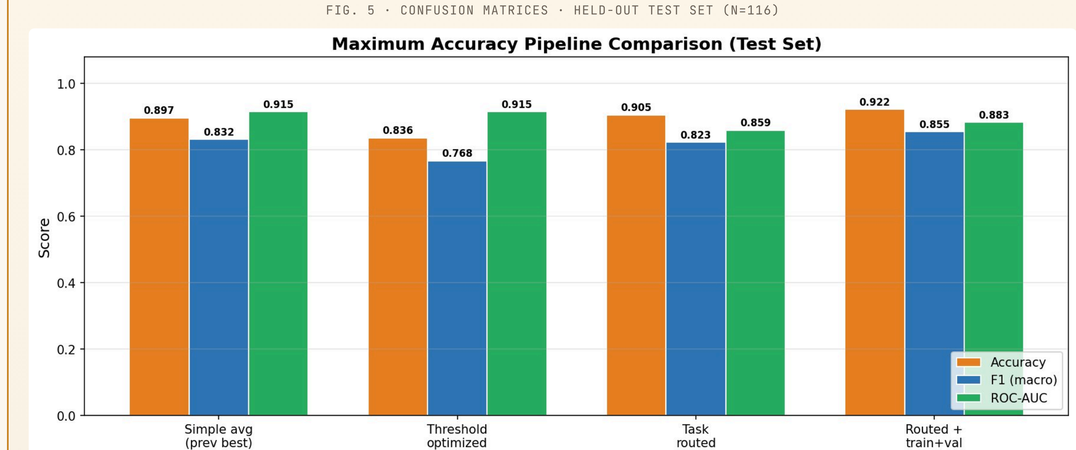


FIG. 6 · PIPELINE COMPARISON: SIMPLE, THRESHOLD, TASK-ROUTED

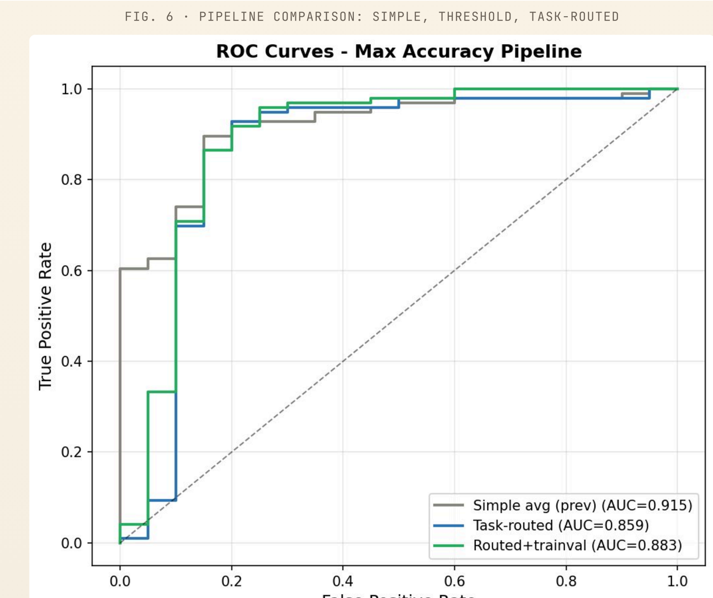


FIG. 7 · ROC CURVES · AUC 0.859–0.915

Trade-off: simple-avg has highest AUC (0.915), but task-routed wins on accuracy (92.2%) and sensitivity (96.9%) at the operating point — under ~1:5 imbalance, catching dementia outweighs uniform separability.

14 Per-prediction transparency

Every prediction is auditable. Below: a **true-positive** dementia case (left) and a **true-negative** control case (right) with full feature attributions.

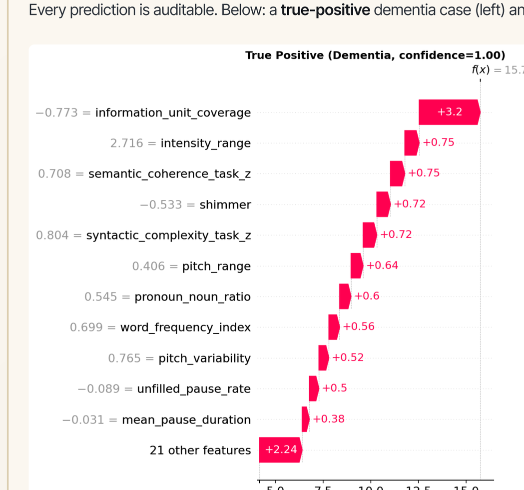


FIG. 8 · TP (DEMENTIA)

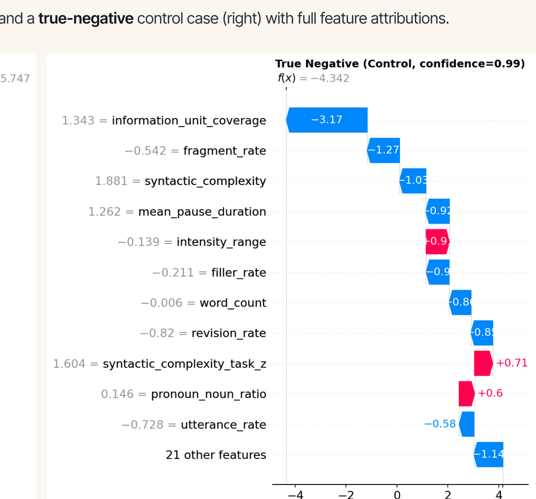


FIG. 9 · TN (CONTROL)

15 Key findings & conclusions

A — Hand-crafted features outperform BERT on small clinical data.
At ~880 training samples, our 32-feature ensemble (92.2%) beats BERT fine-tune (84.5%) by 7.7 pp.

B — Task routing is essential.
Information-unit coverage is the strongest feature on cookie speech but identically zero on 60% of samples. Routing unlocks its signal.

C — Beats SoTA on the same corpus.
92.2% vs. Sarawgi et al. 2020's 88.0% on full Pitt — same split protocol, fairer comparison.

D — Deployable, interpretable, audit-ready.
SHAP traces every prediction back to specific speech behaviours a clinician can inspect.

16 Clinical impact & future directions

The gap this work closes. Existing AD screens require imaging or hour-long batteries administered by trained clinicians. A 60-second speech sample, recorded on a phone, classified in under a second, with a clinician-readable SHAP explanation, makes this the first realistic **population-scale AD pre-screen**.

- **Whisper ASR integration.** Replace manual transcripts with automatic speech recognition for fully end-to-end audio → diagnosis
- **Cookie-specialist BERT-BiLSTM.** Preliminary work suggests ≥94% on cookie-only subset — matches Latif 2025 SoTA on balanced data
- **Longitudinal tracking.** Repeated measurement protocol to detect cognitive decline trajectories rather than snapshot diagnosis
- **Cross-language validation.** Pitt is English-only; replicate on Spanish, French, and Arabic corpora for global deployment
- **Clinical trial integration.** Pharma is actively seeking digital biomarkers for early-AD trial enrolment — immediate B2B path

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